

17 Renormalization of Gauge Theories

We now return to gauge theories, and use the external field formalism described in the previous chapter to study the renormalizability of these theories and to carry out an important calculation.

17.1 The Zinn–Justin Equation

In this section the BRST symmetry described in Section 15.7 will be used to demonstrate a fundamental property of the quantum effective action, first derived by Zinn–Justin [1]. According to the general rules outlined in Section 16.4, the BRST invariance of the action $I[\chi]$ imposes on the effective action $\Gamma[\chi]$ the condition

$$\int d^4x \langle \Delta^n(x) \rangle_{J_\chi} \frac{\delta_L \Gamma[\chi]}{\delta \chi^n(x)} = 0, \quad (17.1.1)$$

where the change in $\chi^n(x)$ under a BRST transformation with infinitesimal fermionic parameter θ is

$$\delta_\theta \chi^n(x) = \theta \Delta^n(x) \quad (17.1.2)$$

and $\langle \cdots \rangle$ here denotes a vacuum expectation value taken in the presence of a current J_χ that makes the vacuum expectation value of the operator fields $X^n(x)$ equal to the c-number functions $\chi^n(x)$. The implied sum over n runs over all of the fields in the BRST formalism; that is, over ω_α , ω_α^* , and h_α as well as the gauge and matter fields that in Section 15.8 we have collectively called ϕ^r . Because $\Delta^n(x)$ is quadratic in the fields when χ^n is a gauge or matter field or ω_α , Eq. (17.1.1) does not in general tell us that the effective action is invariant under the same BRST transformations as the action itself.

To handle this complication, we employ a trick that proves useful in dealing with any sort of nilpotent symmetry transformation. First, we introduce a set of c-number external fields $K_n(x)$, and define a new effective action by

$$\Gamma[\chi, K] \equiv W[J_{\chi, K}, K] - \int d^4x \chi^n(x) J_{\chi, K n}(x), \quad (17.1.3)$$

where the connected vacuum persistence amplitude W is here calculated with the gauge-fixed action¹ $I + \int d^4x \Delta^n K_n$:

$$e^{iW[J, K]} \equiv \int \left[\prod_{n, x} d\chi^n(x) \right] \exp \left(iI + i \int d^4x \Delta^n K_n + i \int d^4x \chi^n J_n \right), \quad (17.1.4)$$

and $J_{\chi, K}$ is the current required to give the fields the expectation values χ in the presence of the external fields K :

$$\left. \frac{\delta_R W[J, K]}{\delta J_n(x)} \right|_{J=J_{\chi, K}} = \chi^n(x). \quad (17.1.5)$$

¹Here I is the action I_{NEW} modified as described in Section 15.7 to depend on the ghost and antighost fields ω_α and ω_α^* and on the auxiliary field h_α , with the subscript ‘NEW’ dropped from now on.

(The K_n must have the same fermionic or bosonic statistics as Δ^n , which is opposite to that of χ^n .) Since the BRST transformation is nilpotent, the quantities $\Delta^n(x)$ are BRST-invariant, so in the same way as in Section 16.4 we can show that the new effective action $\Gamma[\chi, K]$ satisfies a BRST-invariance condition:

$$\int d^4x \langle \Delta^n(x) \rangle_{J,K} \frac{\delta_L \Gamma[\chi, K]}{\delta \chi^n(x)} = 0, \quad (17.1.6)$$

where $\langle \cdots \rangle_{J,K}$ denotes a vacuum expectation value calculated in the presence of the current J and the external fields K :

$$\langle \mathcal{O}[\chi] \rangle_{J,K} \equiv \frac{\int \left[\prod_{n,x} d\chi^n(x) \right] \mathcal{O}[\chi] \exp \left(iI + i \int d^4x \Delta^n K_n + i \int d^4x \chi^n J_n \right)}{\int \left[\prod_{n,x} d\chi^n(x) \right] \exp \left(iI + i \int d^4x \Delta^n K_n + i \int d^4x \chi^n J_n \right)}. \quad (17.1.7)$$

It is convenient to express the expectation value of Δ^n as a variational derivative of the effective action. Taking the right variational derivative of Eq. (17.1.3) with respect to K gives

$$\begin{aligned} \frac{\delta_R \Gamma[\chi, K]}{\delta K_n(x)} &= \left. \frac{\delta_R W[J, K]}{\delta K_n(x)} \right|_{J=J_{\chi,K}} + \int d^4y \left. \frac{\delta_R W[J, K]}{\delta J_m(y)} \right|_{J=J_{\chi,K}} \frac{\delta_R J_{\chi,K} m(y)}{\delta K_n(x)} \\ &\quad - \int d^4y \chi^m(y) \frac{\delta_R J_{\chi,K} m(y)}{\delta K_n(x)}. \end{aligned}$$

Using Eq. (17.1.5) we see that the last two terms cancel, and using the definitions (17.1.4) and (17.1.7) gives us then our desired relation

$$\frac{\delta_R \Gamma[\chi, K]}{\delta K_n(x)} = \left. \frac{\delta_R W[J, K]}{\delta K_n(x)} \right|_{J=J_{\chi,K}} = \langle \Delta^n(x) \rangle_{J_{\chi,K}, K}. \quad (17.1.8)$$

The BRST symmetry condition (17.1.6) may now be written as a simple condition, the Zinn–Justin equation, involving the effective action alone:

$$\int d^4x \frac{\delta_R \Gamma[\chi, K]}{\delta K_n(x)} \frac{\delta_L \Gamma[\chi, K]}{\delta \chi^n(x)} = 0. \quad (17.1.9)$$

As remarked after Eq. (??), the interchange of fields and antifields (or in this case χ^n and K_n) results simply in a change of sign of the left-hand side of Eq. (17.1.9), so this can be written as

$$(\Gamma, \Gamma) = 0, \quad (17.1.10)$$

where the antibracket is calculated here with K_n in place of the antifield of χ^n :

$$\begin{aligned} (F, G) &\equiv \int d^4x \frac{\delta_R F[\chi, K]}{\delta \chi^n(x)} \frac{\delta_L G[\chi, K]}{\delta K_n(x)} \\ &\quad - \int d^4x \frac{\delta_R F[\chi, K]}{\delta K_n(x)} \frac{\delta_L G[\chi, K]}{\delta \chi^n(x)}. \end{aligned} \quad (17.1.11)$$

This is formally the same as the Batalin–Vilkovisky ‘master equation’ discussed in Section 15.9, but appears here as a constraint on the quantum effective action $\Gamma[\chi, K]$ rather than on the fundamental action $S[\chi, \chi^\dagger]$. The Zinn–Justin equation (17.1.10) will be used in the next two sections to show how to renormalize gauge theories, and in Section 22.6 to study anomalies in these theories.

17.2 Renormalization: Direct Analysis

The simplest non-Abelian gauge theories are renormalizable in the ‘Dyson’ sense that the operators in the Lagrangian density all have dimensionality (in powers of mass) four or less. As we saw in Chapter 12, this guarantees that the infinities in the quantum effective action only appear in terms that could be cancelled by the counterterms in interactions of dimensionality four or less. But there is more to renormalizability than this. The Lagrangian density is constrained by gauge invariance and other symmetries. For a theory to be renormalizable, it is necessary that the infinities in the quantum effective action satisfy the same constraints, up to possible renormalizations of the fields.

The effective action $\Gamma[\chi, K]$ is a complicated functional of both χ and K , about which the symmetry condition (17.1.9) says complicated things, but fortunately matters are much simpler for the infinite terms in Γ . We will write the action $S[\chi, K] = I[\chi] + \int d^4x \Delta^n K_n$ as the sum of a term $S_R[\chi, K]$ in which masses and coupling constants are set equal to their renormalized values, plus a correction $S_\infty[\chi, K]$, which contains the counterterms that we intend to cancel the infinities from loop graphs. Both S_R and S_∞ must be taken to have the symmetries of the original action $S[\chi, K]$, so the question is whether the infinite parts of the higher-order contributions to Γ share the same symmetries, so that they can be cancelled by the counterterms in S_∞ .

We may expand Γ in a series of terms Γ_N arising from diagrams with just N loops, plus contributions from graphs with $N - M$ loops (where $1 \leq M \leq N$) involving various counterterms in $S_\infty[\chi, K]$ that will be used to cancel infinities in graphs with a total of M loops:

$$\Gamma[\chi, K] = \sum_{N=0}^{\infty} \Gamma_N[\chi, K]. \quad (17.2.1)$$

The symmetry condition (17.1.10) then reads,² for each N ,

$$\sum_{N'=0}^N (\Gamma_{N'}, \Gamma_{N-N'}) = 0. \quad (17.2.2)$$

In the sum (17.2.1) the leading term is just $\Gamma_0[\chi, K] = S_R[\chi, K]$, which of course is finite. Suppose that, for all $M < N - 1$, all infinities arising from M -loop graphs have been cancelled by counterterms in S_∞ . Then infinities can appear in Eq. (17.2.2) only in the $N' = 0$ and $N' = N$ terms, which are equal, and the infinite part of this condition tells us that the infinite part $\Gamma_{N,\infty}$ of Γ_N is subject to the condition that

$$(S_R, \Gamma_{N,\infty}) = 0. \quad (17.2.3)$$

²Such order-by-order relations may be derived formally by repeating the reasoning of Section 16.1. When the action S_R is replaced with $g^{-1}S_R$, the contribution of a connected L -loop graph with I internal lines and V vertices is multiplied by a factor $g^{I-V} = g^{L-1}$. If the counterterms in S_∞ associated with N -loop diagrams are also provided with factors g^N , then Γ_L is the value for $g = 1$ of the term in Γ of order g^{L-1} . Eq. (17.2.2) then follows by requiring Eq. (17.1.10) to hold in each order in g . In cgs units the action has the same dimensions as $\hbar$, and so appears in the path integral multiplied with a factor $1/\hbar$, so we can also use $\hbar$ as a loop-counting parameter in place of g .

This is a symmetry principle generated by S_R , just like that described by Eqs. (??) and (??). Note in particular that the transformation $X \mapsto (S_R, X)$ acts on the external fields K_n as well as the fields χ^n .

Up to this point, we have used none of the special properties of a renormalizable Yang–Mills theory. Now note that, according to the general power-counting rules of renormalization theory, with all infinities cancelled in subgraphs of Γ_N , the infinite part $\Gamma_{N,\infty}[\chi, K]$ of $\Gamma_N[\chi, K]$ can only be a sum of products of fields (including K) and their derivatives of dimensionality (in powers of mass) four or less. Finally, the arguments of Section 16.4 show that $\Gamma[\chi, K]$ and hence also $\Gamma_{N,\infty}[\chi, K]$ are invariant under all of the *linearly realized* symmetry transformations under which the action is invariant. (As described below, these are: Lorentz transformations, *global* gauge transformations, antighost translations, and the ghost phase transformations associated with ghost number conservation. Of course, the auxiliary fields K_n must be assigned suitable transformation properties under these symmetry transformations.) These conditions together with Eq. (17.2.3) will suffice to tell us all we need to know about the structure of $\Gamma_{N,\infty}[\chi, K]$.

To implement these conditions, we need to know the dimensionalities of the external fields K_n . If a field χ^n has dimensionality d_n (that is, d_n powers of mass), then Δ^n correspondingly has dimensionality $d_n + 1$ (as can be seen by inspection of the BRST transformation rules (??)–(??)), so in order for $\int d^4x K_n \Delta^n$ to be dimensionless, K_n must have dimensionality $3 - d_n$. The fields $A^{\alpha\mu}$, ω^α , and $\omega^{\alpha*}$ all have dimensionalities $d_n = +1$, so the corresponding K_n all have dimensionalities $+2$. (We do not introduce any external field corresponding to h^α , because this field is BRST-invariant.) Any spin 1/2 matter fields ψ_ℓ have dimensionalities $3/2$, and the corresponding K_n thus also have dimensionalities $3/2$. Thus a dimension four quantity like $\Gamma_{N,\infty}[\chi, K]$ is at most quadratic in any of the K_n . Furthermore terms that are of second order in the K_n cannot involve any other fields, except that a term of second order in the K_n for spin 1/2 matter fields may involve at most one additional field of dimensionality unity.

We can now use ghost number conservation to show that in fact $\Gamma_{N,\infty}[\chi, K]$ does not contain any terms of second order in the K_n . For this purpose, we also need the ghost quantum numbers of the K_n . If χ^n has ghost quantum number γ_n , then Δ^n has ghost quantum number $\gamma_n + 1$, so K_n must be assigned ghost quantum number $-\gamma_n - 1$. The ghost quantum numbers of the fields $A^{\alpha\mu}$, ψ^ℓ , ω^α , and $\omega^{\alpha*}$ are respectively 0, 0, $+1$, and -1 , so the corresponding external fields K_n have ghost quantum numbers -1 , -1 , -2 , and 0, respectively. This rules out any terms in $\Gamma_{N,\infty}[\chi, K]$ that are of second order in the K_n , with the one possible exception of a term of second order in the external fields K_α^* associated with $\omega^{\alpha*}$ (and involving no other fields). However, these last terms are also forbidden, for a different reason. The BRST transformation of $\omega^{\alpha*}$ is linear in the fields, with

$$\Delta^{\alpha*} = -h^\alpha, \quad (17.2.4)$$

so here

$$\frac{\delta_L \Gamma_{N,\infty}[\chi, K]}{\delta K_\alpha^*} = \langle \Delta^{\alpha*} \rangle_{J_\chi, K} = -h^\alpha$$

is independent of K_α^* . It follows that $\Gamma_{N,\infty}[\chi, K]$ is linear in $K^{\alpha,*}$, and depends on $K^{\alpha,*}$ only through a term $-\int d^4x K_\alpha^* h^\alpha$. (Both K_α^* and h^α are bosonic, so their order is immaterial.) In particular, for $N > 0$, $\Gamma_{N,\infty}[\chi, K]$ is independent of $K^{\alpha,*}$.

We have seen that $\Gamma_{N,\infty}[\chi, K]$ is at most linear in all of the K_n . We will write it as

$$\Gamma_{N,\infty}[\chi, K] = \Gamma_{N,\infty}[\chi, 0] + \int d^4x \mathcal{D}_N^n[\chi; x] K_n(x). \quad (17.2.5)$$

We also recall that S_R has the K dependence:

$$S_R[\chi, K] = S_R[\chi] + \int d^4x \Delta^n[\chi; x] K_n(x).$$

The terms in (17.2.2) of zeroth and first order in K therefore give³

$$\int d^4x \left[\Delta^n[\chi; x] \frac{\delta_L \Gamma_{N,\infty}[\chi, 0]}{\delta \chi^n(x)} + \mathcal{D}_N^n[\chi; x] \frac{\delta_L S_R[\chi]}{\delta \chi^n(x)} \right] = 0 \quad (17.2.6)$$

and

$$\int d^4x \left[\Delta^n[\chi; x] \frac{\delta_L \mathcal{D}_N^m[\chi; y]}{\delta \chi^n(x)} + \mathcal{D}_N^n[\chi; x] \frac{\delta_L \Delta^m[\chi; y]}{\delta \chi^n(x)} \right] = 0, \quad (17.2.7)$$

respectively. These relations may be made more perspicuous by introducing the quantities

$$\Gamma_N^{(\epsilon)}[\chi] \equiv S_R[\chi] + \epsilon \Gamma_{N,\infty}[\chi, 0], \quad (17.2.8)$$

and

$$\Delta_N^{(\epsilon)n}(x) \equiv \Delta^n(x) + \epsilon \mathcal{D}_N^n(x), \quad (17.2.9)$$

with ϵ infinitesimal. Then (17.2.6) (together with the BRST invariance of S_R) just says that $\Gamma_N^{(\epsilon)}[\chi]$ is invariant under the transformation

$$\chi^n(x) \longrightarrow \chi^n(x) + \theta \Delta_N^{(\epsilon)n}(x), \quad (17.2.10)$$

while Eq. (17.2.7) (together with the nilpotence of the original BRST transformation) tells us that this transformation is *nilpotent*.

We must now consider what form this nilpotent transformation may take. As already mentioned, $\Gamma_{N,\infty}$ consists only of terms of dimensionality four or less, so $\mathcal{D}_N^n$ and hence $\Delta_N^{(\epsilon)n}(x)$ have at most the dimensionality of the original BRST transformation function $\Delta^n(x)$. Also, $\mathcal{D}_N^n$ and hence also $\Delta_N^{(\epsilon)n}(x)$ must have the same Lorentz transformation properties and ghost quantum numbers as $\Delta^n(x)$. Hence the most general form of the transformation (17.2.10) is

$$\begin{aligned} \psi &\longrightarrow \psi + i\theta \omega^\alpha T_\alpha \psi, & A_{\alpha\mu} &\longrightarrow A_{\alpha\mu} + \theta [B_{\alpha\beta} \partial_\mu \omega_\beta + D_{\alpha\beta\gamma} A_{\beta\mu} \omega_\gamma], \\ \omega_\alpha &\longrightarrow \omega_\alpha - \frac{1}{2} \theta E_{\alpha\beta\gamma} \omega_\beta \omega_\gamma, \end{aligned}$$

³The second terms in Eqs. (17.2.6) and (17.2.7) have been put into the form shown here by recalling that, because χ^n and K_n have opposite statistics, for any bosonic functionals A and B ,

$$\frac{\delta_R A}{\delta \chi^n} \frac{\delta_L B}{\delta K_n} = - \frac{\delta_L A}{\delta \chi^n} \frac{\delta_R B}{\delta K_n} = - \frac{\delta_R B}{\delta K_n} \frac{\delta_L A}{\delta \chi^n}.$$

where T_α is some matrix acting on the spinor fields, and $B_{\alpha\beta}$, $D_{\alpha\beta\gamma}$, and $E_{\alpha\beta\gamma}$ are constants, with $E_{\alpha\beta\gamma}$ antisymmetric in β and γ . Also, the transformations of ω_α^* and h_α are linear, and are therefore unchanged:

$$\omega_\alpha^* \longrightarrow \omega_\alpha^* - \theta h_\alpha, \quad h_\alpha \longrightarrow h_\alpha.$$

Next we impose the condition of nilpotence. The most important requirement is that $E_{\alpha\beta\gamma}\omega_\beta\omega_\gamma$ should be invariant. This yields the requirement that $E_{\alpha\beta\gamma}E_{\beta\delta\epsilon}\omega_\delta\omega_\epsilon\omega_\gamma$ should vanish, so that the part of $E_{\alpha\beta\gamma}E_{\beta\delta\epsilon}$ that is totally antisymmetric in δ, ϵ, γ vanishes:

$$E_{\alpha\beta\gamma}E_{\beta\delta\epsilon} + E_{\alpha\beta\epsilon}E_{\beta\gamma\delta} + E_{\alpha\beta\delta}E_{\beta\epsilon\gamma} = 0.$$

But this just tells us that $E_{\alpha\beta\gamma}$ is the structure constant of some Lie algebra $\mathcal{E}$. Because $E_{\alpha\beta\gamma}$ goes to the structure constant $C_{\alpha\beta\gamma}$ of the original gauge Lie algebra $\mathcal{A}$ for $\epsilon \rightarrow 0$, $\mathcal{E}$ must be the same as $\mathcal{A}$, and the structure constants $E_{\alpha\beta\gamma}$ can differ from the original $C_{\alpha\beta\gamma}$ only by a multiplicative factor:⁴

$$E_{\alpha\beta\gamma} = \mathcal{Z}C_{\alpha\beta\gamma}.$$

(This is for simple gauge groups; in the general case we would have a separate factor $\mathcal{Z}$ for each simple subgroup.)

Next we turn to the condition that the transformation (17.2.10) be nilpotent when acting on the gauge fields. The requirement that $B_{\alpha\beta}\partial_\mu\omega_\beta + D_{\alpha\beta\gamma}A_{\beta\mu}\omega_\gamma$ be invariant tells us that

$$D_{\alpha\beta\gamma}D_{\beta\delta\epsilon} - D_{\alpha\beta\epsilon}D_{\beta\delta\gamma} = E_{\beta\epsilon\gamma}D_{\alpha\delta\beta} = \mathcal{Z}C_{\beta\epsilon\gamma}D_{\alpha\delta\beta}$$

and

$$B_{\alpha\beta}E_{\beta\gamma\delta} = D_{\alpha\beta\delta}B_{\beta\gamma}.$$

The first condition has the unique solution⁵

$$D_{\alpha\beta\gamma} = \mathcal{Z}C_{\alpha\beta\gamma}.$$

The second condition tells us that the matrix $B_{\alpha\beta}$ commutes with the adjoint representation of the gauge group, and hence (since we have chosen the structure constants totally antisymmetric) must be proportional to a Kronecker delta, with a coefficient we shall call $\mathcal{ZN}$:

$$B_{\alpha\beta} = \mathcal{ZN}\delta_{\alpha\beta}.$$

Finally, the condition that the transformation (17.2.10) be nilpotent when acting on the fermion fields (if any) requires that $\omega^\alpha T_\alpha\psi$ is invariant. This tells us that

$$[T_\beta, T_\gamma] = iE_{\alpha\beta\gamma}T_\alpha,$$

⁴The requirement of global gauge invariance rules out any non-trivial similarity transformation in the relation between $E_{\alpha\beta\gamma}$ and $C_{\alpha\beta\gamma}$.

⁵The matrix $(D_\gamma)_{\alpha\beta} = D_{\gamma\alpha\beta}/\mathcal{Z}$ satisfies the commutation relations of the gauge Lie algebra, $[D_\gamma, D_\epsilon] = C_{\beta\epsilon\gamma}D_\beta$. But the only representation of a simple Lie algebra with the same dimensionality and $\mathcal{A}$ transformation properties as the adjoint representation of $\mathcal{A}$ is the adjoint representation itself.

so T_α differs from the generator t_α in the original Lagrangian only by a factor $\mathcal{Z}$:

$$T_\alpha = \mathcal{Z}t_\alpha.$$

We have thus seen that, apart from the appearance of the new constants $\mathcal{Z}$ and $\mathcal{N}$, the transformation (17.2.10) is just the BRST transformation with which we started:

$$\psi \longrightarrow \psi + i\mathcal{Z}\theta\omega^\alpha t_\alpha\psi, \quad (17.2.11)$$

$$A_{\alpha\mu} \longrightarrow A_{\alpha\mu} + \mathcal{Z}\theta[\mathcal{N}\partial_\mu\omega_\alpha + C_{\alpha\beta\gamma}A_{\beta\mu}\omega_\gamma], \quad (17.2.12)$$

$$\omega_\alpha \longrightarrow \omega_\alpha - \frac{1}{2}\mathcal{Z}\theta C_{\alpha\beta\gamma}\omega_\beta\omega_\gamma, \quad (17.2.13)$$

$$\omega_\alpha^* \longrightarrow \omega_\alpha^* - \theta h_\alpha, \quad (17.2.14)$$

$$h_\alpha \longrightarrow h_\alpha. \quad (17.2.15)$$

Now we must use this symmetry to constrain the structure of the corrected action (17.2.8). Since this contains only the original renormalized action plus the infinite part of the N -loop contribution, it must be the integral of a Lagrangian density

$$\Gamma_N^{(\epsilon)} = \int d^4x \mathcal{L}_N^{(\epsilon)} \quad (17.2.16)$$

with $\mathcal{L}_N^{(\epsilon)}$ a local function of fields and field derivatives of dimensionality (in powers of mass) no greater than 4. Furthermore, as we found in Section 16.4, $\mathcal{L}_N^{(\epsilon)}$ must be invariant under all the symmetries of the original Lagrangian that act *linearly* on the fields. To identify these symmetries, recall that in generalized ξ -gauge, the ‘new’ Lagrangian density in Eq. (??) takes a form given by replacing the term $-(\partial_\mu A_\alpha^\mu)(\partial_\nu A_\alpha^\nu)/2\xi$ in Eq. (??) with the terms $h_\alpha f_\alpha + \frac{1}{2}\xi h_\alpha h_\alpha$ in Eq. (??):

$$\begin{aligned} \mathcal{L}_{\text{NEW}} = \mathcal{L}_M - \frac{1}{4}F_\alpha^{\mu\nu}F_{\alpha\mu\nu} - \partial_\mu\omega_\alpha^*\partial^\mu\omega_\alpha \\ + C_{\alpha\beta\gamma}(\partial_\mu\omega_\alpha^*)A_\gamma^\mu\omega_\beta + h_\alpha\partial_\mu A_\alpha^\mu + \frac{1}{2}\xi h_\alpha h_\alpha. \end{aligned} \quad (17.2.17)$$

Inspection of this formula reveals the following linear symmetries:

(1) **Lorentz invariance.**

(2) **Global gauge invariance**—that is, invariance under the transformations

$$\delta\psi_\ell(x) = i\epsilon^\alpha(t_\alpha)_\ell^m\psi_m(x), \quad (17.2.18)$$

$$\delta A_\mu^\beta(x) = C_{\beta\gamma\alpha}\epsilon^\alpha A_\mu^\gamma(x), \quad (17.2.19)$$

$$\delta\omega_\beta(x) = C_{\beta\gamma\alpha}\epsilon^\alpha\omega_\gamma(x), \quad (17.2.20)$$

$$\delta\omega_\beta^*(x) = C_{\beta\gamma\alpha}\epsilon^\alpha\omega_\gamma^*(x), \quad (17.2.21)$$

$$\delta h_\beta(x) = C_{\beta\gamma\alpha}\epsilon^\alpha h_\gamma(x), \quad (17.2.22)$$

with constant parameters ϵ^α .

(3) **Antighost translation invariance**—that is, invariance under the transformation

$$\omega_\alpha^*(x) \longrightarrow \omega_\alpha^*(x) + c_\alpha, \quad (17.2.23)$$

with arbitrary constant parameters c_α .

(4) **Ghost number conservation**—that is, the conservation of a ghost number equal to +1 for ω_α , -1 for ω_α^* , and 0 for all other fields.

We shall now proceed to work out the structure of the most general Lagrangian density that is renormalizable, in the sense that it consists only of terms of dimensionality +4 or less, that has these linearly acting symmetries, and that is invariant under the modified BRST transformations (17.2.11)–(17.2.15).

From Eq. (17.2.17) we may conclude that the fields A_α^μ , ω_α , ω_α^* , and h_α have the dimensionalities +1, +1, +1, and +2, respectively (in powers of mass). Note also that ghost number conservation requires that ω and ω^* come in pairs, while antighost translation invariance dictates that ω^* always appears as a derivative. Each pair of ω and $\partial_\mu \omega^*$ fields adds +3 to the dimensionality, so renormalizability rules out any term with more than one such pair. With one such pair we can have at most one more derivative or one additional gauge field, and Lorentz invariance dictates that we must have one or the other. The only renormalizable allowed interactions involving ghost fields are then linear combinations of terms of the form $\partial_\mu \omega_\alpha^* \partial^\mu \omega_\beta$ or $\partial_\mu \omega_\alpha^* A_\gamma^\mu \omega_\beta$.

Next, let us consider the terms that involve the field h_α and possibly other fields but not ω or ω^* . This field has dimensionality +2, so renormalizability and Lorentz invariance allows this field to appear only⁶ multiplied with another h_β or $\partial_\mu A_\beta^\mu$ or $A_\beta^\mu A_{\gamma\mu}$.

Finally, the Lagrangian will contain renormalizable terms involving only the matter and gauge fields. We will call the sum of these terms $\mathcal{L}_{\psi A}$. Putting this all together and using global gauge invariance, the most general renormalizable interaction allowed by the assumed symmetries (aside from BRST invariance) takes the form:

$$\begin{aligned} \mathcal{L}_N^{(\epsilon)} = \mathcal{L}_{\psi A} + \frac{1}{2} \xi' h_\alpha h_\alpha + c h_\alpha \partial_\mu A_\alpha^\mu - e_{\alpha\beta\gamma} h_\alpha A_\beta^\mu A_{\gamma\mu} \\ - Z_\omega (\partial_\mu \omega_\alpha^*) (\partial^\mu \omega_\alpha) - d_{\alpha\beta\gamma} (\partial_\mu \omega_\alpha^*) \omega_\beta A_\gamma^\mu, \end{aligned} \quad (17.2.24)$$

where ξ' , Z_ω , c , $d_{\alpha\beta\gamma}$, and $e_{\alpha\beta\gamma}$ are unknown constants, with no constraints except obvious symmetry properties such as global gauge invariance and $e_{\alpha\beta\gamma} = e_{\alpha\gamma\beta}$. (As mentioned earlier, we are assuming for simplicity that the gauge group is simple, but the extension to a direct sum of simple and $U(1)$ gauge groups would be trivial; for instance, instead of one term proportional to $h_\alpha h_\alpha$ we would have a sum of such terms, one for each simple subgroup of the gauge group.)

Now we impose BRST invariance. The cancellation of terms in $\delta \mathcal{L}_N^{(\epsilon)}$ proportional to $\theta \partial_\mu h_\alpha \partial^\mu \omega_\alpha$ tells us that

$$c = \frac{Z_\omega}{Z\mathcal{N}}. \quad (17.2.25)$$

The cancellation of terms in $\delta \mathcal{L}_N^{(\epsilon)}$ proportional to $\theta \partial_\mu h_\alpha \omega_\beta A_\gamma^\mu$ (or $\theta \partial_\mu \omega_\alpha^* \omega_\beta \partial^\mu \omega_\gamma$) requires that

$$d_{\alpha\beta\gamma} = -(Z_\omega/\mathcal{N}) C_{\alpha\beta\gamma}. \quad (17.2.26)$$

⁶In a theory with scalar fields, we could also have renormalizable terms with h_α multiplied with one or two scalar fields. Such terms cause no trouble, but for brevity they will not be considered here.

The terms in $\delta\mathcal{L}_N^{(\epsilon)}$ proportional to $\theta\partial_\mu\omega_\alpha^*\omega_\beta\omega_\gamma A_\delta^\mu$ automatically then cancel by virtue of the Jacobi identity for the structure constants. The cancellation of terms in $\delta\mathcal{L}_N^{(\epsilon)}$ proportional to $\theta h_\alpha\partial_\mu\omega_\beta A_\gamma^\mu$ (or $\theta h_\alpha A_\beta^\mu A_{\gamma\mu}\omega_\delta$) yields

$$e_{\alpha\beta\gamma} = 0. \quad (17.2.27)$$

Finally, the effect of the infinitesimal transformation (17.2.10) on matter and gauge fields is the same as a local gauge transformation with gauge parameters $\epsilon_\alpha = \mathcal{Z}\mathcal{N}\theta\omega_\alpha$ and gauge couplings renormalized by a factor $1/\mathcal{N}$ (that is, with t_α and $C_{\alpha\beta\gamma}$ replaced with $\tilde{t}_\alpha \equiv t_\alpha/\mathcal{N}$ and $\tilde{C}_{\alpha\beta\gamma} \equiv C_{\alpha\beta\gamma}/\mathcal{N}$), so the cancellation of terms in $\delta\mathcal{L}_N^{(\epsilon)}$ with only one factor of ω_α and no factors h_α or ω_α^* simply tells us that the Lagrangian $\mathcal{L}_{\psi A}$ for these fields is *gauge-invariant*, with a renormalized gauge coupling. We conclude then that the most general renormalizable Lagrangian density allowed by our assumed symmetry principles is

$$\begin{aligned} \mathcal{L}_N^{(\epsilon)} = & -\frac{1}{4}Z_A\tilde{F}_\alpha^{\mu\nu}\tilde{F}_{\alpha\mu\nu} - Z_\psi\bar{\psi}\gamma^\mu[\partial_\mu - i\tilde{t}_\alpha A_{\alpha\mu}]\psi + \frac{1}{2}\xi'h_\alpha h_\alpha \\ & + \frac{Z_\omega}{\mathcal{N}\mathcal{Z}}h_\alpha\partial_\mu A_\alpha^\mu - Z_\omega(\partial_\mu\omega_\alpha^*)(\partial^\mu\omega_\alpha) + Z_\omega\tilde{C}_{\alpha\beta\gamma}(\partial_\mu\omega_\alpha^*)\omega_\beta A_\gamma^\mu, \end{aligned} \quad (17.2.28)$$

where the tilde on $\tilde{F}_\alpha^{\mu\nu}$ indicates that the field strength is to be calculated using the renormalized structure constant $\tilde{C}_{\alpha\beta\gamma} \equiv C_{\alpha\beta\gamma}/\mathcal{N}$. But apart from the appearance of a number of new constant coefficients, this is the same Lagrangian with which we started. The new constants in this Lagrangian (including the gauge coupling constant) may be freely shifted by adjusting the N th-order terms in the corresponding constants in the original unrenormalized Lagrangian. In particular, we can adjust these terms to make $\Gamma_N^{(\epsilon)} = S_R$, in which case $\Gamma_{N,\infty} = 0$, completing the proof.

* * *

In the above proof we made important use of the accidental invariance of the gauge-fixed Lagrangian (17.2.17) under the antighost translation transformation (17.2.23). This symmetry would not be present for gauge-fixing functionals other than $f_\alpha = \partial_\mu A_\alpha^\mu$, which are not spacetime derivatives. One frequently cited example which preserves Lorentz invariance and global gauge invariance is $f_\alpha = \partial_\mu A_\alpha^\mu + a_{\alpha\beta\gamma}A_\beta^\mu A_{\gamma\mu}$, where $a_{\alpha\beta\gamma}$ is a constant matrix, symmetric in β and γ , which transforms as a tensor under global gauge transformations. (Such constant tensors exist for all $SU(N)$ groups with $N \geq 3$.) Another more important example is the background gauge-fixing functional to be introduced in Section 17.4.

The absence of antighost translation invariance does not affect our argument that $\mathcal{L}_N^{(\epsilon)}$ is a Lorentz- and global gauge-invariant local function of fields and field derivatives of dimensionality no greater than 4, that is invariant under the renormalized BRST transformation (17.2.11)–(17.2.15). But without antighost translation invariance there are new terms in $\mathcal{L}_N^{(\epsilon)}$ that satisfy these conditions. Since the transformation (17.2.11)–(17.2.15) is nilpotent, we can construct such terms as $s'F$, where the transformation (17.2.11)–(17.2.15) is written

as $\chi^n \rightarrow \chi^n + \theta s' \chi^n$, and F is an arbitrary Lorentz- and global gauge-invariant function of ghost number -1 . One such term is

$$a_{\alpha\beta\gamma} s'(\omega_\alpha^* A_{\beta\mu} A_\gamma^\mu) = -a_{\alpha\beta\gamma} [h_\alpha A_{\beta\mu} A_\gamma^\mu + 2\mathcal{Z}\omega_\alpha^* (\mathcal{N}\partial_\mu \omega_\beta + C_{\beta\delta\epsilon} A_{\delta\mu} \omega_\epsilon) A_\gamma^\mu].$$

This causes no trouble: it is just a renormalized version of the usual ghost and gauge-fixing terms arising from terms $a_{\alpha\beta\gamma} A_\beta^\mu A_{\gamma\mu}$ in the gauge-fixing functional f_α . But there is also another possible term of the form

$$b_{\alpha\beta\gamma} s'(\omega_\alpha^* \omega_\beta^* \omega_\gamma) = -b_{\alpha\beta\gamma} \left[2h_\alpha \omega_\beta^* \omega_\gamma + \frac{1}{2} \mathcal{Z} C_{\gamma\delta\epsilon} \omega_\alpha^* \omega_\beta^* \omega_\delta \omega_\epsilon \right],$$

where $b_{\alpha\beta\gamma}$ is a constant, antisymmetric in α and β , which transforms as a tensor under global gauge transformations. (Such tensors exist for any Lie group; for instance, we could take $b_{\alpha\beta\gamma}$ to be proportional to $C_{\alpha\beta\gamma}$.) But the Faddeev–Popov–De Witt method cannot yield four-ghost interactions in the Lagrangian, so there are no counterterms available to absorb ultraviolet divergences in this term. This is not just a technical obstacle to proving renormalizability; for gauge-fixing functionals like $f_\alpha = \partial_\mu A_\alpha^\mu + a_{\alpha\beta\gamma} A_\beta^\mu A_{\gamma\mu}$, one-loop graphs actually do yield divergences in four-ghost amplitudes that cannot be cancelled by counterterms in the Faddeev–Popov–De Witt Lagrangian.

Aside from avoiding gauge-fixing functionals other than $f_\alpha = \partial_\mu A_\alpha^\mu$, the only solution to this problem seems to be the one mentioned in Section 15.7. We must give up the Faddeev–Popov–De Witt approach, and instead take the action from the beginning as the most general renormalizable function of the gauge, matter, ghost, and auxiliary fields that is invariant under BRST and the other symmetries of the theory. According to the arguments of Section 15.8, the action can be written in the form $I_0 + s\Psi$ with I_0 ghost-free, and the S -matrix is independent of Ψ , so we can justify this procedure by quantizing the gauge theory in axial gauge, where ghosts decouple, and then taking Ψ to be anything we like. In particular, we can include $s(\omega^* \omega^* \omega)$ terms in the action that can serve as counterterms to the divergences in four-ghost vertices.

17.3 Renormalization: General Gauge Theories⁷

The proof of the renormalizability of non-Abelian gauge theories in the previous section relied on a ‘brute force’ analysis of the possible terms in the action of dimensionality four or less. But as we saw in Chapter 12, this limitation on the dimensionality of terms in the action can be at best a good approximation. The successful renormalizable quantum field theories that are used to describe the strong, weak, and electromagnetic interactions are almost certainly effective field theories, accompanied with terms of dimensionality $d > 4$; these terms are normally not observed because they are suppressed by $4 - d$ powers of some very large mass, perhaps of order 10^{16} – 10^{18} GeV. Gravitation too can be described by an effective field theory, in which the Lagrangian density contains not only the Einstein–Hilbert term $-\sqrt{g}R/16\pi G$, but also all scalars constructed from four or more derivatives

⁷This section lies somewhat out of the book’s main line of development, and may be omitted in a first reading.

of the gravitational field. We need to show that gauge theories of this sort, which are not renormalizable in the power-counting sense, are nonetheless renormalizable in the modern sense that the ultraviolet divergences are governed by the gauge symmetries in such a way that there is a counterterm available to cancel every infinity [3].

For this purpose, let us return to the action $S[\chi, \chi^\dagger]$ introduced in Section 15.9, taken as a function of independent fields χ^n (including gauge and matter fields ϕ^r and ghost fields ω^A as well as the non-minimal fields ω^{*A} and h^A), together with their antifields $\chi_n^\dagger$. In theories like quantum gravity or Yang–Mills theories, that are based on a closed gauge algebra with structure constants f^C_{AB} , this action is constrained to be of the form

$$S = I[\phi] + \omega^A f^r_A[\phi] \phi_r^\dagger + \frac{1}{2} \omega^A \omega^B f^C_{AB}[\phi] \omega_C^\dagger - h^A \omega_A^{*\dagger}, \quad (17.3.1)$$

where $I[\phi]$ is invariant under the infinitesimal gauge transformations $\phi^r \longrightarrow \phi^r + \epsilon^A f^r_A[\phi]$. (As in Section 15.9, the indices r, A , etc., include a spacetime coordinate, over which we integrate in sums over these indices.)

We shall not limit ourselves here to actions of this form, but we shall suppose that the local symmetries of the theory are imposed by requiring that the action must obey some ‘structural constraints’ on its antifield dependence, of which Eq. (17.3.1) provides just one example. The structural constraints are assumed to be linear, in the sense that if they are satisfied for $S + \mathcal{S}_1$ and for $S + \mathcal{S}_2$, then for any constants α_1, α_2 they are satisfied for $S + \alpha_1 \mathcal{S}_1 + \alpha_2 \mathcal{S}_2$. We also impose the quantum master equation (??)

$$(S, S) - 2i\hbar \Delta S = 0, \quad (17.3.2)$$

with ΔS defined by Eq. (??), and the factor $\hbar$ now made explicit as a ‘loop-counting’ parameter, in the sense described in the footnote of the previous section.

The action is taken as a power series in $\hbar$

$$S = S_R + \hbar S_1 + \hbar^2 S_2 + \cdots, \quad (17.3.3)$$

where S_R is an action of the same general form as S , but with all coupling parameters replaced with finite renormalized values, and the S_N are a set of infinite counterterms. The action S is supposed to satisfy the quantum master equation (17.3.2) for all $\hbar$, so S_R satisfies the classical master equation

$$(S_R, S_R) = 0, \quad (17.3.4)$$

while the counterterms satisfy

$$(S_R, S_N) = -\frac{1}{2} \sum_{M=1}^{N-1} (S_M, S_{N-M}) + i\Delta S_{N-1}. \quad (17.3.5)$$

The counterterms S_N are not by themselves sufficient to cancel the ultraviolet divergences in loop graphs. As a generalization of the conventional renormalization of fields, we also have to introduce a set of renormalized fields and antifields, defined in terms of the

original fields and antifields by an arbitrary anticanonical transformation. An infinitesimal anticanonical transformation may be defined in terms of an infinitesimal generating functional δF by Eq. (??), so a sequence of anticanonical transformations $G(t) \longrightarrow G(t + \delta t) = G(t) + (F(t)\delta t, G(t))$ (where $G(t)$ is any functional of fields and antifields, and $F(t)$ is the generating functional) leads to a finite canonical transformation $G \longrightarrow \tilde{G} \equiv G(1)$, with

$$\frac{d}{dt}G(t) = (F(t), G(t)), \quad G(0) = G. \quad (17.3.6)$$

If $F(t)$ is given by a power series

$$F(t) = gtF_1 + \hbar^2 t^2 F_2 + \dots, \quad (17.3.7)$$

then Eqs. (17.3.3), (17.3.6), and (17.3.7) yield a transformed action

$$\begin{aligned} \tilde{S} = S_R + \hbar[S_1 + (F_1, S_R)] \\ + \hbar^2[S_2 + (F_1, S_1) + (F_2, S_R) + \tfrac{1}{2}(F_1, (F_1, S_R))] + \dots \end{aligned} \quad (17.3.8)$$

The question is whether we can use what freedom we have to choose the F_N and S_N so as to cancel all infinities arising from loop graphs.

As already noted, the first term S_R in Eq. (17.3.3) is automatically finite. Suppose that by cancellation of infinities with S_M and F_M for $M < N$ it has been possible to eliminate all infinities in the terms Γ_M of order $\hbar^M$ in the quantum effective action with $M < N$. As we saw in the previous section, the Zinn–Justin equation (derived here by setting $\chi_n^\dagger = K_n + \delta\Psi/\delta\chi^n$) tells us in this case that the infinite part $\Gamma_{N,\infty}$ of the term in the quantum effective action of order $\hbar^N$ satisfies the condition

$$(S_R, \Gamma_{N,\infty}) = 0. \quad (17.3.9)$$

The field and antifield variables χ^n and $\chi_n^\dagger$ are related to the variables χ^n and K_n by an anticanonical transformation, which preserves all antibrackets, so the antibracket in Eq. (17.3.9) may be calculated in terms of χ^n and $\chi_n^\dagger$ instead of χ^n and K_n .

The condition (17.3.5) satisfied by the counterterm S_N is not the same as the condition (17.3.9) satisfied by $\Gamma_{N,\infty}$. However, given any S_N^0 that satisfies Eq. (17.3.5), we can find a class of other solutions

$$S_N = S_N^0 + S'_N, \quad (17.3.10)$$

where S'_N is arbitrary, except for the condition that $S_R + S'_N$, like $S_R + S_N^0$, satisfies the same symmetry conditions as S_R , and that

$$(S_R, S'_N) = 0 \quad (17.3.11)$$

so as not to invalidate Eq. (17.3.5). The infinite part of the N th-order term in the quantum effective action may therefore be written

$$\Gamma_{N,\infty} = S'_{N,\infty} + (F_{N,\infty}, S_R) + X_{N,\infty}, \quad (17.3.12)$$

where X_N consists of terms from loop graphs, as well as from the term S_N^0 and various terms in Γ that involve S_M and F_M for $M < N$. For instance, for $N = 2$ Eq. (17.3.8) gives

$$X_2 = S_2^0 + 2(F_1, S_1) + (F_1, (F_1, S_R)) + \text{two-loop terms involving only } S_R \\ + \text{one-loop terms involving } S_R, S_1 \text{ and } F_1.$$

For our purposes the only thing we need to know about X_N is that it does not involve S'_N or F_N , and that it is invariant under any linearly realized global symmetries of S_R .

Now, because $(S_R, S_R) = 0$, the operation $F \mapsto (S_R, F)$ is nilpotent; for all F ,

$$(S_R, (S_R, F)) = 0. \quad (17.3.13)$$

Hence it follows from Eqs. (17.3.9) and (17.3.11)–(17.3.13) that

$$(S_R, X_{N,\infty}) = 0. \quad (17.3.14)$$

Any term in $X_{N,\infty}$ of the form (S_R, Y) may be cancelled in Eq. (17.3.12) by choosing $F_{N,\infty}$ equal to Y . Thus the space of possible remaining infinite terms in $X_{N,\infty}$ that need to be cancelled by the counterterm $S'_{N,\infty}$ consist of those functionals X that satisfy $(S_R, X) = 0$, counting as equivalent functionals that differ only by terms of the form (S_R, Y) . In other words, the infinities in Γ_N that need to be cancelled by the counterterms S'_N belong to the *cohomology* of the mapping $X \mapsto (S_R, X)$.

The possible form of the counterterm S'_N is limited by the requirement that $S_R + S'_N$ must satisfy whatever structural constraints are imposed on the action S . Thus we can complete the proof of renormalizability if we can show that the cohomology of the mapping $X \mapsto (S_R, X)$ consists only of functionals that satisfy this structural constraint.

In the case of quantum gravity coupled to the Yang–Mills fields of a semisimple gauge symmetry, the symmetries of the theory are implemented by the structural constraint (17.3.1). In this case there is a theorem [4] that states that the cohomology of the mapping $X \mapsto (S_R, X)$ (on the space of local functionals, rather than spacetime-dependent functions, of ghost number zero) consists⁸ of functionals $A[\phi]$ that are invariant under the gauge transformation $\phi^r \rightarrow \phi^r + \epsilon^A f_A^r[\phi]$ with structure constants f^C_{AB} . Any N th-order infinity of this sort may be cancelled with a counterterm S'_N of the same form, so although these theories are not renormalizable in the conventional power-counting sense, they are renormalizable in the sense that all infinities can be eliminated by a choice of parameters in the original bare action $I[\phi]$ and by a suitable renormalization of fields and antifields.

In other theories the cohomology of the map $X \mapsto (S_R, X)$ contains additional terms. This does not necessarily require a weakening of the structural constraints, because the additional terms in the cohomology may not correspond to actual ultraviolet divergences. For instance, in gauge theories with $U(1)$ factors the cohomology contains terms [4] corresponding to a redefinition of the action of the $U(1)$ gauge symmetry on the various fields

⁸Strictly speaking, this is valid if one requires that the constant coefficients in S do not take special values for which S would be invariant under a larger group of local symmetries. For instance, this excludes the case $S = 0$.

of the theory, which if infinite would require us to weaken the structural constraints by leaving the normalization of the transformation functions $f_A^r[\phi]$ in Eq. (17.3.1) arbitrary. But this infinity is forbidden by the same soft-photon theorems that tell us that ratios of $U(1)$ couplings to various fields (like the ratios of the various lepton charges in quantum electrodynamics) are unaffected by radiative corrections. (See Section 10.4.) Where the extra terms in the cohomology do contain ultraviolet divergences, it is necessary to weaken the structural constraint imposed on S in order that there should be a counterterm for every possible ultraviolet divergence. It is not known whether this will always be possible; if not, some theories may have to be rejected because of their unremovable ultraviolet divergences.

17.4 Background Field Gauge

We next turn to a method of calculation that explicitly preserves a sort of gauge invariance, and that therefore proves extremely convenient, especially in one-loop calculations. We consider the effective action $\Gamma[A, \psi, \omega, \omega^*]$ as a functional⁹ of classical external gauge, matter, ghost and antighost fields: $A_{\alpha\mu}(x)$, $\psi_\ell(x)$, $\omega_\alpha(x)$, $\omega_\alpha^*(x)$. Even though ghosts and antighosts never appear in initial or final states, we are considering background ghost and antighost fields as well as gauge and matter fields in order to deal with parts of diagrams that have external ghost or antighost lines.

As described in Section 16.1, $\Gamma[A, \psi, \omega, \omega^*]$ is the sum of connected one-particle-irreducible graphs for the vacuum–vacuum amplitude, calculated in a theory in which the quantum fields A' , ψ' , ω' , ω'^* over which we integrate are replaced in the action with shifted fields $A + A'$, $\psi + \psi'$, $\omega + \omega'$, $\omega^* + \omega'^*$, the path integral being taken over primed fields with the unprimed fields held fixed. We are free to choose the gauge-fixing function $f_\alpha(x)$ pretty much any way we like; instead of our previous choice $f_\alpha = \partial_\mu A_\alpha^\mu$ (or $\partial_\mu[A_\alpha'^\mu + A_\alpha^\mu]$) we shall now take [5]

$$f_\alpha = \partial_\mu A_\alpha'^\mu + C_{\alpha\beta\gamma} A_\beta^\mu A_{\gamma\mu}'. \quad (17.4.1)$$

The reason for this choice is that it makes the gauge-fixing term $f_\alpha f_\alpha$ invariant under a formal transformation, in which the background field A_α^μ transforms as a gauge field, while the quantum field $A_\alpha'^\mu$ transforms homogeneously, like an ordinary matter field that happens to belong to the adjoint representation of the gauge group

$$\delta A_\alpha^\mu = \partial^\mu \epsilon_\alpha - C_{\alpha\beta\gamma} \epsilon_\beta A_\gamma^\mu, \quad (17.4.2)$$

$$\delta A_\alpha'^\mu = -C_{\alpha\beta\gamma} \epsilon_\beta A_\gamma'^\mu. \quad (17.4.3)$$

The transformation properties of f_α can be seen most easily by writing it as a new sort of covariant derivative

$$f_\alpha \equiv \bar{D}_\mu A_\alpha'^\mu, \quad (17.4.4)$$

⁹We are now returning to the specific choice of the gauge-fixing functional $B[f]$ as the Gaussian (??), and we are integrating out the auxiliary field h_α , so that the gauge-fixing term in the modified Lagrangian is just $-f_\alpha f_\alpha/2\xi$.

where for any field ϕ_α in the adjoint representation

$$\bar{D}_\mu \phi_\alpha \equiv \partial_\mu \phi_\alpha + C_{\alpha\beta\gamma} A_{\beta\mu} \phi_\gamma. \quad (17.4.5)$$

We see that under the transformations (17.4.2), (17.4.3), the function (17.4.1) transforms just like A'_α :

$$\delta f_\alpha = -C_{\alpha\beta\gamma} \epsilon_\beta f_\gamma, \quad (17.4.6)$$

so the term $f_\alpha f_\alpha$ in the modified Lagrangian is invariant

$$\delta(f_\alpha f_\alpha) = -C_{\alpha\beta\gamma} f_\alpha \epsilon_\beta f_\gamma = 0. \quad (17.4.7)$$

Also, the original Lagrangian density $\mathcal{L}$ depends on A and A' only through the sum $A + A'$, which under the combined transformation (17.4.2), (17.4.3) undergoes an ordinary gauge transformation

$$\delta(A_\alpha^\mu + A'^\mu_\alpha) = \partial^\mu \epsilon_\alpha - C_{\alpha\beta\gamma} \epsilon_\beta (A_\gamma^\mu + A'^\mu_\gamma). \quad (17.4.8)$$

If we transform the background and quantum matter fields by

$$\delta\psi = it_\alpha \epsilon_\alpha \psi, \quad (17.4.9)$$

$$\delta\psi' = it_\alpha \epsilon_\alpha \psi', \quad (17.4.10)$$

then also

$$\delta(\psi + \psi') = it_\alpha \epsilon_\alpha (\psi + \psi'). \quad (17.4.11)$$

The original Lagrangian $\mathcal{L}$ is invariant under the original gauge transformations (17.4.8), (17.4.11), and only depends on $A + A'$ and $\psi + \psi'$, so it is also invariant under the new formal transformations (17.4.2), (17.4.3), (17.4.9), (17.4.10).

It will be useful to make this invariance property more explicit, by writing $\mathcal{L}$ in terms of the background-covariant derivative $\bar{D}_\mu$. In general, we have

$$\begin{aligned} \mathcal{L} = & -\frac{1}{4} \left(\partial_\mu [A_{\alpha\nu} + A'_{\alpha\nu}] - \partial_\nu [A_{\alpha\mu} + A'_{\alpha\mu}] \right. \\ & \left. + C_{\alpha\beta\gamma} [A_{\beta\mu} + A'_{\beta\mu}] [A_{\gamma\nu} + A'_{\gamma\nu}] \right)^2 \\ & + \mathcal{L}_M \left(\psi + \psi', \partial_\mu (\psi + \psi') - it_\alpha (A_{\alpha\mu} + A'_{\alpha\mu}) (\psi + \psi') \right) \\ = & -\frac{1}{4} \left(F_{\alpha\mu\nu} + \bar{D}_\mu A'_{\alpha\nu} - \bar{D}_\nu A'_{\alpha\mu} + C_{\alpha\beta\gamma} A'_{\beta\mu} A'_{\gamma\nu} \right)^2 \\ & + \mathcal{L}_M \left(\psi + \psi', \bar{D}_\mu (\psi + \psi') - it_\alpha A'_{\alpha\mu} (\psi + \psi') \right), \end{aligned} \quad (17.4.12)$$

where, as in Eq. (17.4.5),

$$\bar{D}_\mu A'_{\alpha\nu} \equiv \partial_\mu A'_{\alpha\nu} + C_{\alpha\beta\gamma} A_{\beta\mu} A'_{\gamma\nu}, \quad (17.4.13)$$

$$\bar{D}_\mu \psi \equiv \partial_\mu \psi - it_\alpha A_{\alpha\mu} \psi, \quad (17.4.14)$$

and $F_{\alpha\mu\nu}$ is the background field strength

$$F_{\alpha\mu\nu} \equiv \partial_\mu A_{\alpha\nu} - \partial_\nu A_{\alpha\mu} + C_{\alpha\beta\gamma} A_{\beta\mu} A_{\gamma\nu}. \quad (17.4.15)$$

(The square in the first term of $\mathcal{L}$ is intended to imply obvious index contractions.) Clearly $\mathcal{L}$ is invariant under the new transformations (17.4.2), (17.4.3), (17.4.9), (17.4.10), because it involves $A_{\alpha\mu}$ only in the field strength $F_{\alpha\mu\nu}$ and in background covariant derivatives $\bar{D}_\mu$ of ‘matter’ fields $A'_{\alpha\mu}$, ψ' and ψ .

This new transformation should be carefully distinguished from a true gauge transformation. Such a transformation can have no effect on A or ψ , which are just prescribed classical background fields, and induces an ordinary gauge transformation on $A + A'$ and $\psi + \psi'$, so

$$\delta_{\text{TRUE}} A'_\alpha = 0, \quad (17.4.16)$$

$$\begin{aligned} \delta_{\text{TRUE}} A'^\mu_\alpha &= \partial^\mu \epsilon_\alpha - C_{\alpha\beta\gamma} \epsilon_\beta (A^\mu_\gamma + A'^\mu_\gamma) \\ &= \bar{D}^\mu \epsilon_\alpha - C_{\alpha\beta\gamma} \epsilon_\beta A'^\mu_\gamma, \end{aligned} \quad (17.4.17)$$

and

$$\delta_{\text{TRUE}} \psi = 0, \quad (17.4.18)$$

$$\delta_{\text{TRUE}} \psi' = i t_\alpha \epsilon_\alpha (\psi + \psi'). \quad (17.4.19)$$

Of course, this is the same as the formal transformations (17.4.2), (17.4.3), (17.4.9), (17.4.10) in its effect on $A + A'$ and $\psi + \psi'$, and therefore also leaves the original Lagrangian $\mathcal{L}$ invariant. However, for our new choice (17.4.1) of f_α , the term $f_\alpha f_\alpha$ does not depend only on $A + A'$, and is not invariant under (17.4.16) and (17.4.17). Instead,

$$\delta_{\text{TRUE}} f_\alpha = \bar{D}_\mu (\bar{D}^\mu \epsilon_\alpha - C_{\alpha\beta\gamma} \epsilon_\beta A'^\mu_\gamma) \quad (17.4.20)$$

with $\bar{D}_\mu$ given by (17.4.5).

Finally, let us consider the ghost Lagrangian in this new gauge. The quantity (??) in the ghost action is given in general by just replacing ϵ_α with the ghost field $\omega_\alpha + \omega'_\alpha$ in $\delta_{\text{TRUE}} f_\alpha$:

$$\Delta_\alpha = \bar{D}_\mu [\bar{D}^\mu (\omega_\alpha + \omega'_\alpha) - C_{\alpha\beta\gamma} (\omega_\beta + \omega'_\beta) A'^\mu_\gamma]. \quad (17.4.21)$$

The ghost Lagrangian in Eq. (??) is therefore

$$\mathcal{L}_{\text{GH}} = (\omega_\alpha^* + \omega'^*_\alpha) \bar{D}_\mu [\bar{D}^\mu (\omega_\alpha + \omega'_\alpha) - C_{\alpha\beta\gamma} (\omega_\beta + \omega'_\beta) A'^\mu_\gamma] \quad (17.4.22)$$

or, integrating by parts,

$$\mathcal{L}_{\text{GH}} = - \left(\bar{D}_\mu (\omega_\alpha^* + \omega'^*_\alpha) \right) \left(\bar{D}^\mu (\omega_\alpha + \omega'_\alpha) - C_{\alpha\beta\gamma} (\omega_\beta + \omega'_\beta) A'^\mu_\gamma \right). \quad (17.4.23)$$

This is manifestly invariant under the joint transformations (17.4.2), (17.4.3), supplemented now with transformations on ω and ω' :

$$\delta \omega_\alpha = -C_{\alpha\beta\gamma} \epsilon_\beta \omega_\gamma, \quad (17.4.24)$$

$$\delta \omega'_\alpha = -C_{\alpha\beta\gamma} \epsilon_\beta \omega'_\gamma, \quad (17.4.25)$$

and likewise

$$\delta \omega_\alpha^* = -C_{\alpha\beta\gamma} \epsilon_\beta \omega_\gamma^*, \quad (17.4.26)$$

$$\delta\omega'_\alpha{}^* = -C_{\alpha\beta\gamma}\epsilon_\beta\omega'_\gamma{}^*. \quad (17.4.27)$$

We see that the formal combined transformations (17.4.2), (17.4.3), (17.4.9), (17.4.10), and (17.4.24)–(17.4.27) leave invariant the complete Lagrangian in the modified action (??):

$$\mathcal{L}_{\text{MOD}} = \mathcal{L} - \frac{1}{2\xi}f_\alpha f_\alpha + \mathcal{L}_{\text{GH}}. \quad (17.4.28)$$

We are integrating over A' , ψ' , ω' , and ω'^* with a measure that is presumed to be invariant under the simple matrix transformations (17.4.3), (17.4.10), (17.4.25), and (17.4.27), so the effective action $\Gamma[A, \psi, \omega, \omega^*]$ is invariant under the remaining transformations (17.4.2), (17.4.9), (17.4.24), and (17.4.26). In other words, it is gauge-invariant in the same sense as the original action $I[A, \psi, \omega, \omega^*]$.

This formal gauge invariance sets powerful constraints on the infinities that can occur in the effective action. The ultraviolet divergences in Γ appear in the coefficients of terms whose dimensionalities are $[\text{mass}]^d$ with $d \leq 4$, but here these terms are invariant under the background gauge transformations (17.4.2), (17.4.9), (17.4.24), and (17.4.26). For instance, in a gauge theory based on a simple gauge group, with spin 1/2 fermions belonging to an irreducible representation of this group, the only such terms are of the form

$$\Gamma_\infty = \int d^4x \mathcal{L}_\infty, \quad (17.4.29)$$

$$\begin{aligned} \mathcal{L}_\infty = & -\frac{1}{4}L_A F_{\alpha\mu\nu}F_\alpha{}^{\mu\nu} - L_\psi \bar{\psi}\gamma^\mu \bar{D}_\mu\psi \\ & - mL_m \bar{\psi}\psi - L_\omega(\bar{D}_\mu\omega_\alpha^*)(\bar{D}^\mu\omega_\alpha), \end{aligned} \quad (17.4.30)$$

where here $F_{\alpha\mu\nu}$, $\bar{D}_\mu\psi$, $\bar{D}_\mu\omega_\alpha$, and $\bar{D}_\mu\omega_\alpha^*$ are constructed entirely from background fields.¹⁰

$$F_{\alpha\mu\nu} \equiv \partial_\mu A_{\alpha\nu} - \partial_\nu A_{\alpha\mu} + C_{\alpha\beta\gamma}A_{\beta\mu}A_{\gamma\nu}, \quad (17.4.31)$$

$$\bar{D}_\mu\psi \equiv \partial_\mu\psi - it_\alpha A_{\alpha\mu}\psi, \quad (17.4.32)$$

$$\bar{D}_\mu\omega_\alpha \equiv \partial_\mu\omega_\alpha + C_{\alpha\beta\gamma}A_{\beta\mu}\omega_\gamma, \quad (17.4.33)$$

$$\bar{D}_\mu\omega_\alpha^* \equiv \partial_\mu\omega_\alpha^* + C_{\alpha\beta\gamma}A_{\beta\mu}\omega_\gamma^*. \quad (17.4.34)$$

Dimensional analysis leads us to expect that the constants L_A , L_ψ , L_m , and L_ω are logarithmically divergent.

To deal with these infinities, we note that the Lagrangian (17.4.12) contains a purely classical piece

$$\mathcal{L}_{\text{CLASS}} = -\frac{1}{4}F_{\alpha\mu\nu}F_\alpha{}^{\mu\nu} - \bar{\psi}(\gamma^\mu \bar{D}_\mu + m)\psi - (\bar{D}_\mu\omega_\alpha^*)(\bar{D}^\mu\omega_\alpha) \quad (17.4.35)$$

¹⁰The condition of a simple gauge group insures that there is just a single kinetic term for A and for ω , proportional to $F_{\alpha\mu\nu}F_\alpha{}^{\mu\nu}$ and $(\bar{D}_\mu\omega_\alpha^*)(\bar{D}^\mu\omega_\alpha)$ respectively, while the condition that ψ transforms irreducibly insures that there is just a single kinetic term and a single mass term for ψ . It would be easy to treat more general possibilities, at the cost of a slight complication in notation. We are also implicitly using the conservation of ghost number.

obtained by dropping all terms in $\mathcal{L}_{\text{MOD}}$ that involve the quantum fields A' , ψ' , ω' , ω'^* . We define renormalized fields

$$A_{\alpha\mu}^R \equiv \sqrt{1 + L_A} A_{\alpha\mu}, \quad (17.4.36)$$

$$\psi_\ell^R \equiv \sqrt{1 + L_\psi} \psi_\ell, \quad (17.4.37)$$

$$\omega_\alpha^R \equiv \sqrt{1 + L_\omega} \omega_\alpha, \quad (17.4.38)$$

$$\omega_\alpha^{R*} \equiv \sqrt{1 + L_\omega} \omega_\alpha^*, \quad (17.4.39)$$

so that the sum of the terms (17.4.30) and (17.4.35) takes the form

$$\begin{aligned} \mathcal{L}_{\text{CLASS}} + \mathcal{L}_\infty = & -\frac{1}{4} F_{\alpha\mu\nu}^R F_\alpha^{R\mu\nu} - \bar{\psi}^R \gamma^\mu D_\mu^R \psi^R \\ & - m^R \bar{\psi}^R \psi^R - (D_\mu^R \omega_\alpha^{*R})(D^{\mu R} \omega_\alpha^R), \end{aligned} \quad (17.4.40)$$

where m^R is the renormalized mass

$$m^R \equiv m(1 + L_m)/(1 + L_\psi), \quad (17.4.41)$$

and

$$F_{\alpha\mu\nu}^R \equiv \partial_\mu A_{\alpha\nu}^R - \partial_\nu A_{\alpha\mu}^R + C_{\alpha\beta\gamma}^R A_{\beta\mu}^R A_{\gamma\nu}^R, \quad (17.4.42)$$

$$D_\mu^R \psi^R \equiv \partial_\mu \psi^R - i t_\alpha^R A_{\alpha\mu}^R \psi^R, \quad (17.4.43)$$

$$D_\mu^R \omega_\alpha^R \equiv \partial_\mu \omega_\alpha^R + C_{\alpha\beta\gamma}^R A_{\beta\mu}^R \omega_\gamma^R, \quad (17.4.44)$$

$$D_\mu^R \omega_\alpha^{*R} \equiv \partial_\mu \omega_\alpha^{*R} + C_{\alpha\beta\gamma}^R A_{\beta\mu}^R \omega_\gamma^{*R}. \quad (17.4.45)$$

The renormalized structure constants and group generators here are just

$$C_{\alpha\beta\gamma}^R \equiv (1 + L_A)^{-1/2} C_{\alpha\beta\gamma}, \quad (17.4.46)$$

$$t_\alpha^R \equiv (1 + L_A)^{-1/2} t_\alpha. \quad (17.4.47)$$

Because we assumed that the Lie algebra here is simple, the structure constant $C_{\alpha\beta\gamma}$ and group generator t_α are fixed by the group structure except for a single common factor, the unrenormalized gauge coupling constant g . Eqs. (17.4.46) and (17.4.47) thus simply tell us that the gauge coupling constant g^R factor in C^R and t^R is renormalized by

$$g^R = g(1 + L_A)^{-1/2}. \quad (17.4.48)$$

This result exhibits the particular virtue of the background field gauge. In a general gauge we would encounter independent renormalization factors for the gauge field and the gauge coupling constant, and we would have to calculate two separate amplitudes (say, the vacuum polarization and three-gauge-field vertex function) in order to be able to sort these out. In background field gauge, background field gauge invariance ties these two renormalizations together by requiring that the infinite terms in the effective Lagrangian involve the field strength in its original form (17.4.31), and so we can calculate the charge renormalization factor by studying just one gauge field amplitude.

17.5 A One-Loop Calculation in Background Field Gauge

As an exercise, we are now going to calculate the one-loop renormalization factor for the gauge coupling constant in a general non-Abelian gauge theory. As we will see in the next chapter, this provides an essential input in so-called ‘renormalization group’ calculations of physical processes at high energy; the results we obtain here will be used there to demonstrate the asymptotic freedom of non-Abelian gauge theories.

The method to be employed here is somewhat novel. Usually, one considers the effective action in a spacetime-dependent background gauge field, and calculates the terms quadratic in this field, extracting a factor $(q_\mu A_{\alpha\nu} - q_\nu A_{\alpha\mu})^2$ (where q is the gauge field four-momentum), and only then isolating the logarithmic divergence by setting $q = 0$ in the coefficient of this factor. Instead we shall follow the much simpler course of taking the gauge field to be spacetime-independent from the beginning. In this case the terms in the effective action that are quadratic or cubic in the gauge field of course vanish, but there is a non-vanishing quartic term that is ultraviolet divergent, and which can be used to calculate the coupling constant renormalization factor $(1 + L_A)^{-1/2}$. In this way our one-loop calculation becomes a matter of simple matrix algebra. Note that this procedure can only work in background field gauge; otherwise there would be independent logarithmic divergences in the parts of the effective action that are quartic and quadratic in the background gauge field.

With this motivation, we turn to the calculation of the one-loop effective action in a background field for which $A_{\alpha\mu}$ is constant and $\psi = \omega = \omega^* = 0$. For such a background field, the full modified Lagrangian is¹¹

$$\mathcal{L}_{\text{MOD}} = \mathcal{L} + \mathcal{L}_f + \mathcal{L}_{GH}, \quad (17.5.1)$$

$$\begin{aligned} \mathcal{L} = & -\frac{1}{4} (F_{\alpha\mu\nu} + \bar{D}_\mu A'_{\alpha\nu} - \bar{D}_\nu A'_{\alpha\mu} + C_{\alpha\beta\gamma} A'_{\beta\mu} A'_{\gamma\nu})^2 \\ & - \bar{\psi}' (\not{D} - it_\alpha A'_\alpha + m) \psi', \end{aligned} \quad (17.5.2)$$

$$\mathcal{L}_f = -\frac{1}{2\xi} f_\alpha f_\alpha = -\frac{1}{2\xi} (\bar{D}_\mu A'^\mu_\alpha)^2, \quad (17.5.3)$$

$$\mathcal{L}_{GH} = -(\bar{D}_\mu \omega'^*_\alpha) (\bar{D}^\mu \omega'_\alpha - C_{\alpha\beta\gamma} \omega'_\beta A'^\mu_\gamma). \quad (17.5.4)$$

One-loop graphs for vacuum–vacuum amplitudes are calculated from the part of the action that is quadratic in the quantum fields $A', \psi', \omega', \omega'^*$ over which one integrates. Keeping only such quadratic terms, we have

$$\begin{aligned} \mathcal{L}_{\text{QUAD}} = & -\frac{1}{4} (\bar{D}_\mu A'_{\alpha\nu} - \bar{D}_\nu A'_{\alpha\mu})^2 - \frac{1}{2} F_\alpha^{\mu\nu} C_{\alpha\beta\gamma} A'_{\beta\mu} A'_{\gamma\nu} \\ & - \bar{\psi}' (\not{D} + m) \psi' - \frac{1}{2\xi} (\bar{D}_\mu A'^\mu_\alpha)^2 - (\bar{D}_\mu \omega'^*_\alpha) (\bar{D}^\mu \omega'_\alpha). \end{aligned} \quad (17.5.5)$$

¹¹See Eqs. (17.4.12), (17.4.4), and (17.4.23). We are here specializing to the case of matter fields forming a multiplet of spin 1/2 fermions. The squares in $\mathcal{L}$ and $\mathcal{L}_f$ include obvious index contractions.

The corresponding action may be put in the general quadratic form:

$$\begin{aligned}
I_{\text{QUAD}} &\equiv \int d^4x \mathcal{L}_{\text{QUAD}} \\
&= -\frac{1}{2} \int d^4x d^4y A'^\mu_\alpha(x) A'^\nu_\beta(y) \mathcal{D}^A_{x\alpha\mu, y\beta\nu} - \int d^4x d^4y \bar{\psi}'_k(x) \psi'_\ell(y) \mathcal{D}^\psi_{xk, y\ell} \\
&\quad - \int d^4x d^4y \omega'^*_\alpha(x) \omega'_\beta(y) \mathcal{D}^\omega_{x\alpha, y\beta},
\end{aligned} \tag{17.5.6}$$

with

$$\begin{aligned}
\mathcal{D}^A_{x\alpha\mu, y\beta\nu} &\equiv \eta_{\mu\nu} \left(-\delta_{\gamma\alpha} \frac{\partial}{\partial x_\lambda} + C_{\gamma\delta\alpha} A^\lambda_\delta(x) \right) \left(-\delta_{\gamma\beta} \frac{\partial}{\partial y_\lambda} + C_{\gamma\epsilon\beta} A_{\epsilon\lambda}(y) \right) \delta^4(x-y) \\
&\quad - \left(-\delta_{\gamma\alpha} \frac{\partial}{\partial x^\nu} + C_{\gamma\delta\alpha} A_{\delta\nu}(x) \right) \left(-\delta_{\gamma\beta} \frac{\partial}{\partial y^\mu} + C_{\gamma\epsilon\beta} A^\mu_\epsilon(y) \right) \delta^4(x-y) \\
&\quad + F_{\gamma\mu\nu}(x) C_{\gamma\alpha\beta} \delta^4(x-y) \\
&\quad + \frac{1}{\xi} \left(-\delta_{\gamma\alpha} \frac{\partial}{\partial x^\mu} + C_{\gamma\delta\alpha} A_{\delta\mu}(x) \right) \left(-\delta_{\gamma\beta} \frac{\partial}{\partial y^\nu} + C_{\gamma\epsilon\beta} A_{\epsilon\nu}(y) \right) \delta^4(x-y),
\end{aligned} \tag{17.5.7}$$

$$\mathcal{D}^\psi_{xk, y\ell} \equiv \left(-\gamma^\mu \frac{\partial}{\partial y^\mu} - it_\alpha A_\alpha(y) + m \right)_{k\ell} \delta^4(x-y), \tag{17.5.8}$$

$$\mathcal{D}^\omega_{x\alpha, y\beta} \equiv \left(-\delta_{\gamma\alpha} \frac{\partial}{\partial x^\lambda} + C_{\gamma\delta\alpha} A_{\delta\lambda}(x) \right) \left(-\delta_{\gamma\beta} \frac{\partial}{\partial y_\lambda} + C_{\gamma\epsilon\beta} A^\lambda_\epsilon(y) \right) \delta^4(x-y). \tag{17.5.9}$$

(The minus signs in front of $\partial/\partial x$ and $\partial/\partial y$ drop out when we integrate by parts.)

The one-loop contribution to the effective action is given (as in Section 16.2) by

$$\begin{aligned}
\exp(i\Gamma^{(1\text{ loop})}[A]) &\propto \int_{1\text{PI}} \left(\prod dA' \right) \left(\prod d\psi' \right) \left(\prod d\bar{\psi}' \right) \left(\prod d\omega' \right) \left(\prod d\omega'^* \right) \\
&\quad \times \exp(iI_{\text{QUAD}}[A', \psi', \bar{\psi}', \omega', \omega'^*; A]) \\
&\propto (\text{Det } \mathcal{D}^A)^{-1/2} (\text{Det } \mathcal{D}^\psi)^{+1} (\text{Det } \mathcal{D}^\omega)^{+1}.
\end{aligned} \tag{17.5.10}$$

(The exponents $-1/2$ and $+1$ appear because A' is a real boson field, while ψ' , $\bar{\psi}'$, ω' , and ω'^* are distinct fermionic fields.) The calculation of such determinants is generally not an easy task. However, it becomes much simpler in the case of constant external fields, where the $\mathcal{D}$ s can be diagonalized by passing to momentum space.

Let us therefore now consider the case of constant background field $A_{\alpha\mu}$. For non-Abelian gauge theories such a constant field *cannot* be removed by a gauge transformation, as shown by the non-zero values of various gauge-covariant fields

$$F_{\alpha\mu\nu} = C_{\alpha\beta\gamma} A_{\beta\mu} A_{\gamma\nu}, \tag{17.5.11}$$

$$D_\lambda F_{\delta\mu\nu} = C_{\delta\epsilon\alpha} C_{\alpha\beta\gamma} A_{\epsilon\lambda} A_{\beta\mu} A_{\gamma\nu}, \tag{17.5.12}$$

and so on. Lorentz and gauge invariance tell us how to express the part of $\Gamma[A]$ of a given dimensionality as the integral of a finite number of local functions of $F_{\alpha\mu\nu}$, $D_\lambda F_{\alpha\mu\nu}$, etc.; the

coefficients of the terms in this expression can be inferred by comparing the contribution that these terms make to $\Gamma[A]$ for constant background field $A_{\alpha\mu}$ with the results of a perturbative expansion.

We transform each of the ‘matrices’ $\mathcal{D}^A$, $\mathcal{D}^\psi$, and $\mathcal{D}^\omega$ to a momentum basis by the usual normalized Fourier transform

$$\mathcal{D}_{q\cdots,p\cdots} = \int \frac{d^4x}{(2\pi)^2} e^{-iq\cdot x} \int \frac{d^4y}{(2\pi)^2} e^{ip\cdot y} \mathcal{D}_{x\cdots,y\cdots} \quad (17.5.13)$$

With A constant, this gives

$$\mathcal{D}_{q\cdots,p\cdots} = \delta^4(p - q) \mathcal{M}_{\cdots,\cdots}(q), \quad (17.5.14)$$

where $\cdots$ denotes discrete indices, and the $\mathcal{M}$ are finite q -dependent matrices

$$\begin{aligned} \mathcal{M}_{\alpha\mu,\beta\nu}^A(q) = & \eta_{\mu\nu}(-iq^\lambda\delta_{\gamma\alpha} + A_\delta^\lambda C_{\gamma\delta\alpha})(iq_\lambda\delta_{\gamma\beta} + A_{\epsilon\lambda}C_{\gamma\epsilon\beta}) \\ & - (-iq_\nu\delta_{\gamma\alpha} + A_{\delta\nu}C_{\gamma\delta\alpha})(iq_\mu\delta_{\gamma\beta} + A_\epsilon^\mu C_{\gamma\epsilon\beta}) + F_{\gamma\mu\nu}C_{\gamma\alpha\beta} \\ & + \frac{1}{\xi}(-iq_\mu\delta_{\gamma\alpha} + A_{\delta\mu}C_{\gamma\delta\alpha})(iq_\nu\delta_{\gamma\beta} + A_{\epsilon\nu}C_{\gamma\epsilon\beta}) + \epsilon \text{ terms}, \end{aligned} \quad (17.5.15)$$

$$\mathcal{M}_{k\ell}^\psi(q) = (i\not{q} - it_\alpha \not{A}_\alpha + m)_{k\ell} + \epsilon \text{ terms}, \quad (17.5.16)$$

$$\mathcal{M}_{\alpha\beta}^\omega(q) = (-iq_\lambda\delta_{\gamma\alpha} + A_{\delta\lambda}C_{\gamma\delta\alpha})(iq^\lambda\delta_{\gamma\beta} + A_\epsilon^\lambda C_{\gamma\epsilon\beta}) + \epsilon \text{ terms}, \quad (17.5.17)$$

with $F_{\gamma\mu\nu}$ given by Eq. (17.5.11). From Eq. (17.5.10) we have then

$$\begin{aligned} i\Gamma^{(1 \text{ loop})}[A] = & -\frac{1}{2} \ln \text{Det } \mathcal{D}^A + \ln \text{Det } \mathcal{D}^\psi + \ln \text{Det } \mathcal{D}^\omega \\ = & -\frac{1}{2} \text{Tr} \ln \mathcal{D}^A + \text{Tr} \ln \mathcal{D}^\psi + \text{Tr} \ln \mathcal{D}^\omega \\ = & \delta^4(p - p) \int d^4q \left[-\frac{1}{2} \text{tr} \ln \mathcal{M}^A(q) + \text{tr} \ln \mathcal{M}^\psi(q) + \text{tr} \ln \mathcal{M}^\omega(q) \right]. \end{aligned} \quad (17.5.18)$$

We denote traces by ‘tr’ instead of ‘Tr’ in the last line of Eq. (17.5.18) (and from now on in this section) to indicate that these are the usual traces of finite matrices rather than of integral operators.

Since we are aiming here at a calculation of the infinite factor L_A multiplying FF terms in the effective action, let’s isolate the term in (17.5.18) that is of fourth order in the background field A . For this purpose, it is convenient to divide each $\mathcal{M}$ into terms $\mathcal{M}_n$ containing $n = 0, 1$, or 2 factors of A :

$$\mathcal{M} = \mathcal{M}_0 + \mathcal{M}_1 + \mathcal{M}_2. \quad (17.5.19)$$

It is then elementary algebra to show that the term in (17.5.18) of fourth order in $A_{\alpha\mu}$ is

$$[\text{tr} \ln \mathcal{M}]_{A^4} = \text{tr} \left\{ -\frac{1}{2} [\mathcal{M}_0^{-1} \mathcal{M}_2]^2 + [\mathcal{M}_0^{-1} \mathcal{M}_1]^2 \mathcal{M}_0^{-1} \mathcal{M}_2 - \frac{1}{4} [\mathcal{M}_0^{-1} \mathcal{M}_1]^4 \right\}. \quad (17.5.20)$$

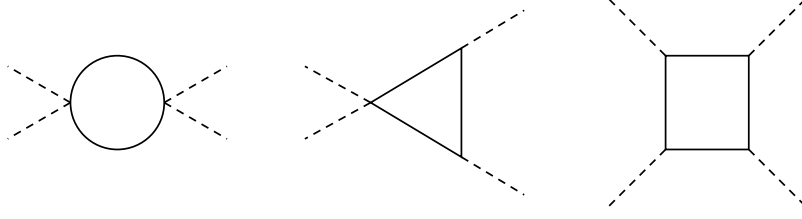


Figure 17.1. One-loop Feynman diagrams for the term in the quantum effective action that is quartic in a constant background gauge field A_α^μ . Here solid lines represent internal gauge, ghost, or matter lines; dashed lines indicate factors of A_α^μ . These three diagrams correspond to the three terms in Eq. (17.5.20).

(To see this, insert factors ϵ and ϵ^2 multiplying $\mathcal{M}_1$ and $\mathcal{M}_2$ in Eq. (17.5.19), differentiate $\text{tr} \ln \mathcal{M}$ four times with respect to ϵ , divide by $4!$ and set $\epsilon = 0$.) The $\mathcal{M}_0^{-1}$ factors here are just the usual propagators; for $\xi = 1$, these are

$$[\mathcal{M}_0^A(q)]_{\alpha\mu,\beta\nu}^{-1} = \delta_{\alpha\beta} \eta_{\mu\nu} (q^2 - i\epsilon)^{-1}, \quad (17.5.21)$$

$$[\mathcal{M}_0^\psi(q)]_{k\ell}^{-1} = [i\not{q} + m]_{k\ell}^{-1}, \quad (17.5.22)$$

$$[\mathcal{M}_0^\omega(q)]_{\alpha,\beta}^{-1} = \delta_{\alpha\beta} (q^2 - i\epsilon)^{-1}. \quad (17.5.23)$$

Indeed, the three terms in Eq. (17.5.20) just correspond to the three Feynman diagrams shown in Figure 17.1; the present method of calculation saves us from having to think about signs and combinatoric factors.

For the A loop, Eq. (17.5.15) gives for $\xi = 1$:

$$\begin{aligned} [\mathcal{M}_1^A(q)]_{\alpha\mu,\beta\nu} &= -2\eta_{\mu\nu} q^\lambda [\mathcal{A}_\lambda]_{\alpha\beta}, \\ [\mathcal{M}_2^A(q)]_{\alpha\mu,\beta\nu} &= \left[\eta_{\mu\nu} \mathcal{A}_\lambda \mathcal{A}^\lambda - \mathcal{A}_\nu \mathcal{A}_\mu - \mathcal{A}_\mu \mathcal{A}_\nu \right]_{\alpha\beta} + F_{\gamma\mu\nu} C_{\gamma\alpha\beta}, \end{aligned}$$

where $\mathcal{A}_\lambda$ is the matrix

$$[\mathcal{A}_\lambda]_{\alpha\beta} \equiv -iC_{\alpha\beta\gamma} A_{\gamma\lambda}$$

for which

$$\begin{aligned} [\mathcal{A}_\lambda, \mathcal{A}_\rho]_{\alpha\beta} &= -C_{\alpha\delta\gamma} C_{\delta\beta\epsilon} (A_{\gamma\lambda} A_{\epsilon\rho} - A_{\gamma\rho} A_{\epsilon\lambda}) \\ &= -(C_{\alpha\delta\gamma} C_{\delta\beta\epsilon} + C_{\alpha\delta\epsilon} C_{\delta\beta\gamma}) A_{\gamma\lambda} A_{\epsilon\rho} \\ &= +C_{\alpha\delta\beta} C_{\delta\epsilon\gamma} A_{\gamma\lambda} A_{\epsilon\rho} = C_{\alpha\delta\beta} F_{\delta\rho\lambda}. \end{aligned}$$

The integrals have the structure

$$\begin{aligned} \int d^4q q^\mu q^\nu f(q^2) &= \frac{1}{4} \eta^{\mu\nu} \int d^4q q^2 f(q^2), \\ \int d^4q q^\mu q^\nu q^\rho q^\sigma f(q^2) &= \frac{1}{24} [\eta^{\mu\nu} \eta^{\rho\sigma} + \eta^{\mu\rho} \eta^{\nu\sigma} + \eta^{\mu\sigma} \eta^{\nu\rho}] \int d^4q (q^2)^2 f(q^2). \end{aligned}$$

We then find, for $\xi = 1$:

$$\begin{aligned}
& \int d^4 q \operatorname{tr} \left\{ [\mathcal{M}_0^A(q)^{-1} \mathcal{M}_2^A(q)]^2 \right\} \\
& \quad = 4\mathcal{I} \operatorname{tr} [\mathcal{A}^\lambda \mathcal{A}_\lambda \mathcal{A}^\eta \mathcal{A}_\eta] + 4\mathcal{I} C_{\gamma\alpha\beta} C_{\delta\alpha\beta} F_{\gamma\mu\nu} F_\delta^{\mu\nu}, \\
& \int d^4 q \operatorname{tr} \left\{ [\mathcal{M}_0^A(q)^{-1} \mathcal{M}_1^A(q)]^2 \mathcal{M}_0^A(q)^{-1} \mathcal{M}_2^A(q) \right\} \\
& \quad = 4\mathcal{I} \operatorname{tr} [\mathcal{A}^\lambda \mathcal{A}_\lambda \mathcal{A}^\eta \mathcal{A}_\eta], \\
& \int d^4 q \operatorname{tr} \left\{ [\mathcal{M}_0^A(q)^{-1} \mathcal{M}_1^A(q)]^4 \right\} \\
& \quad = \frac{8}{3} \mathcal{I} \operatorname{tr} \left[\begin{array}{c} 2\mathcal{A}^\lambda \mathcal{A}_\lambda \mathcal{A}^\eta \mathcal{A}_\eta \\ + \mathcal{A}^\lambda \mathcal{A}^\eta \mathcal{A}_\lambda \mathcal{A}_\eta \end{array} \right],
\end{aligned}$$

where $\mathcal{I}$ is the divergent integral

$$\mathcal{I} \equiv \int d^4 q [q^2 - i\epsilon]^{-2}, \quad (17.5.24)$$

whose significance is discussed below. Putting this together in Eq. (17.5.20), we have

$$\begin{aligned}
\int d^4 q [\operatorname{tr} \ln \mathcal{M}^A(q)]_{A^4} &= \frac{2}{3} \mathcal{I} \operatorname{tr} [\mathcal{A}^\lambda \mathcal{A}_\lambda \mathcal{A}^\eta \mathcal{A}_\eta - \mathcal{A}^\lambda \mathcal{A}^\eta \mathcal{A}_\lambda \mathcal{A}_\eta] \\
&\quad - 2\mathcal{I} C_{\gamma\alpha\beta} C_{\delta\alpha\beta} F_{\gamma\mu\nu} F_\delta^{\mu\nu}.
\end{aligned}$$

Both terms are actually of the same form, and when combined yield

$$\int d^4 q [\operatorname{tr} \ln \mathcal{M}^A(q)]_{A^4} = -\frac{5}{3} \mathcal{I} C_{\gamma\alpha\beta} C_{\delta\alpha\beta} F_{\gamma\mu\nu} F_\delta^{\mu\nu}. \quad (17.5.25)$$

Now jumping ahead to the ghost loop, we see from Eq. (17.5.17) that

$$[\mathcal{M}_1^\omega(q)]_{\alpha\beta} = -2[\mathcal{A}^\lambda]_{\alpha\beta} q_\lambda, \quad (17.5.26)$$

$$[\mathcal{M}_2^\omega(q)]_{\alpha\beta} = [\mathcal{A}^\lambda \mathcal{A}_\lambda]_{\alpha\beta}. \quad (17.5.27)$$

We then find

$$\begin{aligned}
& \int d^4 q \operatorname{tr} \left\{ [\mathcal{M}_0^\omega(q)^{-1} \mathcal{M}_2^\omega(q)]^2 \right\} \\
& \quad = \mathcal{I} \operatorname{tr} [\mathcal{A}^\lambda \mathcal{A}_\lambda \mathcal{A}^\eta \mathcal{A}_\eta], \\
& \int d^4 q \operatorname{tr} \left\{ [\mathcal{M}_0^\omega(q)^{-1} \mathcal{M}_1^\omega(q)]^2 \mathcal{M}_0^\omega(q)^{-1} \mathcal{M}_2^\omega(q) \right\} \\
& \quad = \mathcal{I} \operatorname{tr} [\mathcal{A}^\lambda \mathcal{A}_\lambda \mathcal{A}^\eta \mathcal{A}_\eta], \\
& \int d^4 q \operatorname{tr} \left\{ [\mathcal{M}_0^\omega(q)^{-1} \mathcal{M}_1^\omega(q)]^4 \right\} \\
& \quad = \frac{2}{3} \mathcal{I} \operatorname{tr} \left[\begin{array}{c} 2\mathcal{A}^\lambda \mathcal{A}_\lambda \mathcal{A}^\eta \mathcal{A}_\eta \\ + \mathcal{A}^\lambda \mathcal{A}^\eta \mathcal{A}_\lambda \mathcal{A}_\eta \end{array} \right].
\end{aligned}$$

Thus for the ghost loop the integral of the quantity (17.5.20) is

$$\begin{aligned}\int d^4q [\text{Tr} \ln \mathcal{M}^\omega(q)]_{A^4} &= \frac{1}{6} \mathcal{I} \text{Tr}[\mathcal{A}^\lambda \mathcal{A}_\lambda \mathcal{A}^\eta \mathcal{A}_\eta - \mathcal{A}^\lambda \mathcal{A}^\eta \mathcal{A}_\lambda \mathcal{A}_\eta] \\ &= \frac{1}{12} \mathcal{I} C_{\gamma\alpha\beta} C_{\delta\alpha\beta} F_{\gamma\mu\nu} F_\delta^{\mu\nu}.\end{aligned}\quad (17.5.28)$$

Finally, the vertices in the matter loop are

$$[\mathcal{M}_1^\psi(q)]_{k\ell} = -i(t_\alpha \mathbb{A}_\alpha)_{k\ell}, \quad [\mathcal{M}_2^\psi(q)]_{k\ell} = 0,$$

so here there is only one term in Eq. (17.5.20):

$$\int d^4q [\text{Tr} \ln \mathcal{M}^\psi(q)]_{A^4} = -\frac{1}{4} \int d^4q \text{Tr} \left\{ [(i\not{q} + m)^{-1} t_\alpha \mathbb{A}_\alpha]^4 \right\}.$$

We are interested in the ultraviolet-divergent part of this integral, so we may drop the mass (which is negligible for large q^ν) and write

$$\begin{aligned}\int d^4q [\text{Tr} \ln \mathcal{M}^\psi(q)]_{A^4} &= -\frac{1}{4} \text{Tr}\{t_\alpha t_\beta t_\gamma t_\delta\} A_{\alpha\mu} A_{\beta\nu} A_{\gamma\rho} A_{\delta\sigma} \\ &\quad \times \int d^4q \frac{\text{Tr}\{\not{q}\gamma^\mu \not{q}\gamma^\nu \not{q}\gamma^\rho \not{q}\gamma^\sigma\}}{(q^2 - i\epsilon)^8} \\ &= -\frac{\mathcal{I}}{96} \text{Tr}\{t_\alpha t_\beta t_\gamma t_\delta\} A_{\alpha\mu} A_{\beta\nu} A_{\gamma\rho} A_{\delta\sigma} \\ &\quad \times \text{Tr} \left\{ \begin{aligned} &2\gamma_\lambda \gamma^\mu \gamma^\lambda \gamma^\nu \gamma_\eta \gamma^\rho \gamma^\eta \gamma^\sigma \\ &+ \gamma_\lambda \gamma^\mu \gamma_\eta \gamma^\nu \gamma^\lambda \gamma^\rho \gamma^\eta \gamma^\sigma \end{aligned} \right\}.\end{aligned}\quad (17.5.29)$$

where $\mathcal{I}$ is the same divergent integral as in Eq. (17.5.24). To calculate the traces of Dirac matrices, we use the anticommutation relations of these matrices to write

$$\begin{aligned}\text{Tr} \left\{ \begin{aligned} &2\gamma_\lambda \gamma^\mu \gamma^\lambda \gamma^\nu \gamma_\eta \gamma^\rho \gamma^\eta \gamma^\sigma \\ &+ \gamma_\lambda \gamma^\mu \gamma_\eta \gamma^\nu \gamma^\lambda \gamma^\rho \gamma^\eta \gamma^\sigma \end{aligned} \right\} &= 8 \text{Tr}\{\gamma^\mu \gamma^\nu \gamma^\rho \gamma^\sigma\} \\ &\quad - 4 \text{Tr}\{\gamma^\nu \gamma^\mu \gamma^\rho \gamma^\sigma\} - 4 \text{Tr}\{\gamma^\mu \gamma^\rho \gamma^\nu \gamma^\sigma\} \\ &= -64\eta^{\mu\rho}\eta^{\nu\sigma} + 32\eta^{\mu\nu}\eta^{\rho\sigma} + 32\eta^{\mu\sigma}\eta^{\nu\rho}.\end{aligned}$$

Eq. (17.5.29) then gives

$$\begin{aligned}\int d^4q [\text{Tr} \ln \mathcal{M}^\psi(q)]_{A^4} &= \frac{1}{3} \mathcal{I} \text{Tr}\{[t_\alpha, t_\beta][t_\gamma, t_\delta]\} A_{\alpha\mu} A_{\beta\nu} A_\gamma^\mu A_\delta^\nu \\ &= -\frac{1}{3} \mathcal{I} F_{\gamma\mu\nu} F_\delta^{\mu\nu} \text{Tr}\{t_\gamma, t_\delta\}.\end{aligned}\quad (17.5.30)$$

Using Eqs. (17.5.25), (17.5.28), and (17.5.30) in Eq. (17.5.18) gives at last

$$\Gamma_{A^4}^{(1 \text{ loop})} = \frac{-i\mathcal{I}}{(2\pi)^4} \int d^4x F_{\gamma\mu\nu} F_\delta^{\mu\nu} \left[\left(\frac{5}{6} + \frac{1}{12} \right) C_{\gamma\alpha\beta} C_{\delta\alpha\beta} - \frac{1}{3} \text{Tr}\{t_\gamma, t_\delta\} \right], \quad (17.5.31)$$

where we have expressed the momentum-space delta-function in (17.5.14) as

$$\delta^4(p - p) = (2\pi)^{-4} \int d^4x 1. \quad (17.5.32)$$

It is important that the result turns out to depend on $A_{\alpha\mu}$ only through the field strength (17.5.11), as required by background gauge invariance.

Let us now (for the first time in this section) use the assumed simplicity of the gauge group and irreducibility of the matter field multiplet. In this case

$$C_{\gamma\alpha\beta}C_{\delta\alpha\beta} = g^2 C_1 \delta_{\gamma\delta}, \quad (17.5.33)$$

$$\text{Tr}\{t_\gamma, t_\delta\} = g^2 C_2 \delta_{\gamma\delta}, \quad (17.5.34)$$

where g is the common gauge coupling constant appearing as a factor in $C_{\gamma\alpha\beta}$ and t_γ , and C_1 and C_2 are numerical constants that characterize the gauge group and the representation of this group provided by the matter multiplet. For instance, in the original Yang–Mills theory the gauge group is $SU(2)$ (or equivalently $SO(3)$) and the structure constants are

$$C_{\gamma\alpha\beta} = g\epsilon_{\gamma\alpha\beta}$$

with α, β , and γ running over the values 1, 2, 3. Comparing with Eq. (17.5.33), we see that here

$$C_1 = 2.$$

Also in this theory the matter field forms a doublet with t_α given by $g/2$ times the usual Pauli matrix σ_α , so

$$C_2 = 1/2.$$

Somewhat more generally, for the group $SU(N)$ with n_f fermions in the defining representation, with a conventional normalization of generators, we have¹²

$$C_1 = N, \quad C_2 = n_f/2. \quad (17.5.35)$$

Returning now to the general case, Eqs. (17.5.33) and (17.5.34) give

$$\Gamma_{A^4}^{(1 \text{ loop})} = \frac{-ig^2 \mathcal{I}}{(2\pi)^4} \int d^4x F_{\gamma\mu\nu} F_\gamma^{\mu\nu} \left[\frac{11}{12} C_1 - \frac{1}{3} C_2 \right]. \quad (17.5.36)$$

That is, the infinite constant L_A in Eq. (17.4.30) is

$$L_A = \frac{4ig^2 \mathcal{I}}{(2\pi)^4} \left(\frac{11}{12} C_1 - \frac{1}{3} C_2 \right). \quad (17.5.37)$$

It remains to say a word about the interpretation of the divergent integral $\mathcal{I}$. First, before we try to integrate over the three-momentum $\mathbf{q}$, we can rotate the contour of integration of q^0 in Eq. (17.5.24) to the imaginary axis; as usual, the $-i\epsilon$ in the denominator forces us to rotate counterclockwise, so that $q^0 = iq^4$, with q^4 running from $-\infty$ to $+\infty$. The integral is then

$$\mathcal{I} = i \int_0^\infty \frac{2\pi^2 q^3 dq}{q^4}, \quad (17.5.38)$$

¹²For $SU(3)$ the t_α are taken as $g/2$ times the Gell-Mann matrices λ_α used in Section 19.7, so that $C_{\alpha\beta\gamma} = (g/2)f_{\alpha\beta\gamma}$.

where q is now the magnitude of the Euclidean four-vector (q^1, q^2, q^3, q^4) . To go further, we evidently need some method of regulating the integral. The simplest way of dealing with the ultraviolet divergence is just to cut off the integral for q above a scale Λ . However, we also need a lower cut-off to deal with the infrared divergence. This is provided by the physics of the situation. If the momenta of the four vector particles is not zero, then a momentum flows through the internal lines of the diagrams, providing an infrared cut-off at the scale μ of these momenta. Similarly, if we evaluate the fourth variational derivative of $\Gamma[A]$ with respect to A not at $A = 0$ but at a finite A , then the propagators of the internal lines do not blow up at zero momenta, and we have an infrared cut-off at a scale $\mu \sim gA$. Either way, $\mathcal{I}$ takes the form

$$\mathcal{I} = 2\pi^2 i \int_{\mu}^{\Lambda} \frac{dq}{q} = 2\pi^2 i \ln \left(\frac{\Lambda}{\mu} \right) \quad (17.5.39)$$

and hence

$$L_A = -\frac{g^2}{2\pi^2} \left(\frac{11}{12}C_1 - \frac{1}{3}C_2 \right) \ln \left(\frac{\Lambda}{\mu} \right) + O(g^4). \quad (17.5.40)$$

Eq. (17.4.48) then gives the renormalized coupling as

$$g_R = g \left[1 + \frac{g^2}{4\pi^2} \ln \left(\frac{\Lambda}{\mu} \right) \left(\frac{11}{12}C_1 - \frac{1}{3}C_2 \right) + O(g^4) \right]. \quad (17.5.41)$$

We note that while in quantum electrodynamics the radiative corrections discussed in Section 11.2 decrease the physical coupling g_R relative to the bare coupling g , in non-Abelian gauge theories they *increase* the physical coupling over the bare coupling, provided that the fermion multiplet is small enough so that $C_2 < 11C_1/4$. The importance of this point will be explored in Chapter 18.

Alternatively, we can deal with the ultraviolet divergence by the methods of dimensional regularization, discussed in Section 11.2. Here, in place of Eq. (17.5.39), we write

$$\mathcal{I} = i \int_0^{\infty} \frac{2\pi^2 q^{d-1} dq}{(q^2 + \mu^2)^2}, \quad (17.5.42)$$

where d is a complex dimensionality, allowed to approach 4 at the end of the calculation, and μ is an infrared cut-off, again taken of the order of the external momenta (or of the background fields times g). As long as d is complex with $\text{Re } d < 0$ and $\mu^2 > 0$, this has the finite value

$$\mathcal{I} = -i\pi^2 \left(\frac{d}{2} - 1 \right) \mu^{d-4} \frac{\pi}{\sin[(d/2 - 2)\pi]}.$$

Analytically continuing to $d \rightarrow 4$, this is

$$\mathcal{I} \longrightarrow -2i\pi^2 \left[\frac{1}{d-4} + \ln \mu + \dots \right], \quad (17.5.43)$$

where $\dots$ denotes finite μ -independent terms. Here we have

$$L_A = \frac{g^2}{2\pi^2} \left(\frac{11}{12}C_1 - \frac{1}{3}C_2 \right) \left(\frac{1}{d-4} + \ln \mu + \dots \right) + O(g^4)$$

and so

$$g_R = g \left[1 - \frac{g^2}{4\pi^2} \left(\frac{11}{12} C_1 - \frac{1}{3} C_2 \right) \left(\frac{1}{d-4} + \ln \mu + \cdots \right) + O(g^4) \right]. \quad (17.5.44)$$

Note that the ultraviolet divergence here takes a different form, but the dependence on the infrared cut-off μ is the same. Eq. (17.5.44) will provide an important input to our discussion of asymptotic freedom in Section 18.7.

Problems

1. Carry out the proof of renormalizability given in Section 17.2, including elementary scalar fields in the Lagrangian.
2. Carry out the quantization of a non-Abelian gauge theory in background field gauge, using the BRST method of quantization discussed at the end of Section 17.2.
3. Derive the relation (17.5.44) between the renormalized and unrenormalized gauge couplings by calculating the terms in $\Gamma^{(1 \text{ loop})}$ that are quadratic in a spacetime-dependent gauge field.
4. Calculate the one-loop relation between the renormalized and unrenormalized gauge couplings in a gauge theory containing elementary scalar fields.

References

1. J. Zinn-Justin, in *Trends in Elementary Particle Theory International Summer Institute on Theoretical Physics in Bonn 1974* (Springer-Verlag, Berlin, 1975).
2. Before the advent of BRST symmetry, proofs of the renormalizability of non-Abelian gauge theories were based directly on the Slavnov–Taylor identities for gauge transformations; see B. W. Lee and J. Zinn-Justin, *Phys. Rev. D* **5**, 3121, 3137 (1972); *Phys. Rev. D* **7**, 1049 (1972); G. 't Hooft and M. Veltman, *Nucl. Phys. B* **50**, 318 (1972); B. W. Lee, *Phys. Rev. D* **9**, 933 (1974). The original proof of renormalizability based on BRST symmetry was given by C. Becchi, A. Rouet, and R. Stora, *Commun. Math. Phys.* **42**, 127 (1975); in *Renormalization Theory—Proceedings of the International School of Mathematical Physics at Erice, August 1975*, eds. G. Velo and A.S. Wightman (D. Reidel, Dordrecht, 1976): pp. 269–97, 299–343. The proof given here follows the general outline of J. Zinn-Justin, Ref. 1; B. W. Lee, in *Methods in Field Theory*, eds. R. Balian and J. Zinn-Justin (North-Holland, Amsterdam, 1976): pp. 79–139.
3. The point of view and presentation here is based on that of J. Gomis and S. Weinberg, Kyoto-Texas preprint RIMS-1036/UTTG-18-95 (1995), to be published in *Nuclear Physics B*. For earlier use of these methods, see B. L. Voronov and I. V. Tyutin, *Theor. Math. Phys.* **50**, 218 (1982); **52**, 628 (1982); B. L. Voronov, P. M. Lavrov, and I. V. Tyutin, *Sov. J. Nucl. Phys.* **36**, 292 (1982); P. M. Lavrov and I. V. Tyutin *Sov. J. Nucl. Phys.* **41**, 1049 (1985); D. Anselmi, *Class. and Quant. Grav.* **11**, 2181

- (1994); **12**, 319 (1995); M. Harada, T. Kugo, and K. Yamawaki, *Prog. Theor. Phys.* **91**, 801 (1994).
4. G. Barnich and M. Henneaux, *Phys. Rev. Lett.* **72**, 1588 (1994); G. Barnich, F. Brandt, and M. Henneaux, *Phys. Rev.* **51**, R143 (1995); *Commun. Math. Phys.* **174**, 57, 93 (1995); *Nucl. Phys. B***455**, 357 (1995).
 5. The background field gauge was introduced by B. S. De Witt, *Phys. Rev.* **162**, 1195, 1239 (1967). For the treatment of multi-loop effects, see G. 't Hooft, in *Functional and Probabilistic Methods in Quantum Field Theory: Proceedings of the 12th Karpacz Winter School of Theoretical Physics* (Acta Universitatis Wratislavenensis no. 38, 1975); B. S. De Witt, in *Quantum Gravity II*, eds. C. Isham, R. Penrose, and D. Sciama (Oxford University Press, Oxford, 1982); L. F. Abbott, *Nucl. Phys. B***185**, 189 (1981).